UNIVERSITY OF CHINESE ACADEMY OF SCIENCES

DOCTORAL THESIS

The measurement of Higgs production cross section in the four-lepton final state at CMS

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Probing four lepton final-state for Higgs Cross section measurement

1.1 Introduction

The ATLAS and CMS collaborations of CERN announced the discovery of much awaited Higgs boson in early July 2012 [1, 2] verifying the underlying concept of electroweak symmetry breaking (EWSB) in the SM of particle physics. This triggered essential precise studies of this particle to confirm its nature and to discover some new physics in case of some deviation while studying its properties. Later on, these experiments have also studied its comprehensive properties [3, 4, 5, 6] which are found to be consistent with the SM Higgs. There is a legacy of measurements on this topic from CMS and ATLAS collaborations in different LHC data taking eras for different decay cahannels of Higgs which can been studied in Ref. [**Khachatryan:2015yvw**, **CMS:2015hja**, **CMS:2016rqf**, **Sirunyan:2017exp**, **CMS:2018hhg**, **atlas-conf-2019-029**, **atlas-conf-2019-025** , **atlas-conf-2019-032**, **CMS:2019drq**, 7, 8, 9, 10, 11, 12, 13]. However, these measurements can be improved with newer data at higher luminosities and improved statistical techniques.

The cross section measurements (integrated and differential) are performed using collisions data of CMS (LHC) corresponding to 137.1 fb^{-1} integrated luminosity at $\sqrt{s} = 13$ TeV. The studied differential observables are sensitive to production mechanism of Higgs. Further details follow in next sections.To minimize the model dependence, results are extracted in the fiducial volume that closely matches with the detector geometry as shown in Figure 1.1 [**ref_fiducial**]. Conribution of other model dependence is extracted by repeating the measurements for SM and other exotic models. The differences are reported as systematic uncertainty. All measurements includes the corrections for the finite detection efficiency and resolution effects, and are extracted assuming $m_H = 125.09$ GeV, as measured from $H \to 2\gamma$ and $H \to 4\ell$ decay modes at CMS experiment of LHC [14].

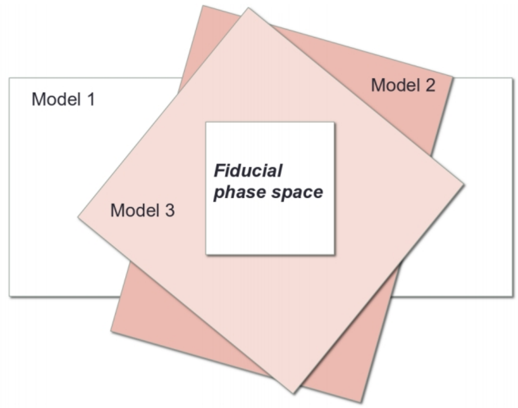


FIGURE 1.1: Fiducial phase definition among different models.

1.2 Datasets and Triggers

The analysis uses full Run 2 (2016, 2017 and 2018) CMS data corresponding to $137.1\ \mathrm{fb}^{-1}$ intergrated luminosity.

Used datasets are listed and documented in **[CMS:2019chr]** with the corresponding integrated luminosities. The analysis relies on four different primary datasets (PDs), *DoubleMuon*, *MuEG*, *EGamma*, and *SingleMuon*, each of which combines a certain collections of HLT paths. Events are collected to avoid duplications from different PDs, using:

- EGamma if they pass the diEle or triEle or singleElectron triggers,
- DoubleMuon if they pass the diMuon or triMuon triggers and fail the diEle and triEle triggers,
- MuEG if they pass the MuEle or MuDiEle or DiMuEle triggers and fail the diEle, triEle, singleElectron, diMuon and triMuon triggers,
- SingleMuon if they pass the singleMuon trigger and fail all the above triggers.

The HLT paths used for full Run 2 collision data is listed in **[CMS:2019chr]** and also in the Appendix.

1.2.1 Trigger Efficiency

Efficiency in data of combination of triggers used for the anlysis is measured by considering 4ℓ events triggered by single lepton triggers. Among the reconstructed four leptons, is geometrically associated with an object that passes at least one single electron or muon triggers. Rest of three leptons are regarded as “probes". In a 4ℓ event, the possible tag-probe pairs are 4 and are counted in denominator while computing efficiency. For each of three leptons which are probes, all matching trigger filter objects are collected. Then the matched trigger filter objects of the three probe leptons are combined in attempt to reconstruct any of the triggers used in the anlysis. If any of the analysis triggers can be formed using the probe leptons, probes are call passing probes which are countedas numerator for the efficiency measurement.

This technique is used to measure efficiency on MC and the comparison with data is exploited to estimate the reliability of simulation. Fig. 1.2 shows the measured efficiency both in data and MC (2018) varied with minimum p_{T} of leptons. MC efficiency seems to describe data well within the uncertainties.

1.3 Simulation Samples

Simulations of the interesting events help us build a reallistic model of experiment in high energy physics. It enables us for a detailed study of physics processes (signal and background). In high energy physics experiments, different MC event generators are used to simulate different process. For this analysis, in particular, descriptions of the production of SM Higgs through process of gluon-gluon fusion (Fig. 1.3) are acquired using POWHEG V2 [15, 16] generator. At accuracy level of next-to-next-to-leading order (i.e. NNLO) in perturbative QCD (pQCD), $\mathrm{gg} \to \mathrm{H}$ process description is obtained by partonic level MC generator NNLOPS.

The POWHEG, provides perturbative QCD calculations at next-to-leading (NLO) accuracy with a connection to parton-shower programs. Using the POWHEG, Higgs production via $\mathrm{gg} \to \mathrm{H}$ with zero associated jets can be described at accuracy level

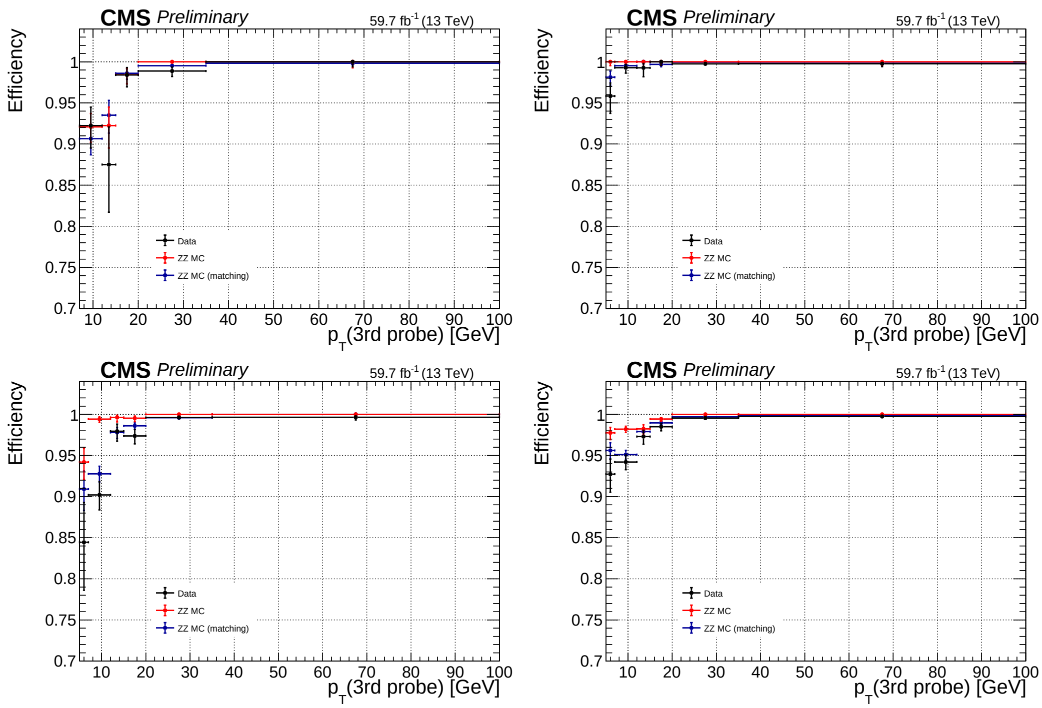


FIGURE 1.2: Measured trigger efficiency in 2018 data where the event collection is made using single electron and muon triggers: $4e$ (top left), 4μ (top right), $2e2\mu$ (bottom left) and 4ℓ (bottom right) final states.

of NLO. Finite masses of top and bottom quarks has been considered for $\mathrm{gg} \to \mathrm{H}$ production for all generators.

Signal Samples

Descriptions of the SM Higgs boson production are obtained using the POWHEG V2 [17, 18, 19] generator for the five main production modes (ggH, VBF, WH, ZH and ttH) as described in section **??** [20], [21], [22], [23]. However, WH and ZH production modes are simulated by the MiNLO HVJ extension of the POWHEG [23]. Higgs decay to four lepton is simulated by JHUGEN generator [24]. For the WH, ZH and ttH modes, Higgs is allowed to decay to H$\to ZZ \to 2\ell 2$X such that two leptons from 4-lepton events are produced from the decay of associated W, Z bosons or top quarks respectively. Showering of parton-level events is done using PYTHIA8.209, and in all cases matching is performed by allowing QCD emissions at all energies in the shower and vetoing them afterwards according to the POWHEG internal scale. All samples are generated with the default NNPDF 3.1

NLO parton distribution functions (PDFs) [25]. The list of signal samples with the corresponding cross sections are shown in **[CMS:2019chr]**.

These samples, for each production mode, are also used to asses model dependence of measured results as shown in the Table **??**. The details of the method used to asses the model dependence will be described in Section 1.5. The POWHEG and NNLOPS based samples for $\mathrm{gg} \rightarrow \mathrm{H}$ production are utilized as theoretical predictions based on SM calculations for the comparison with measured results as described in in Section **??**.

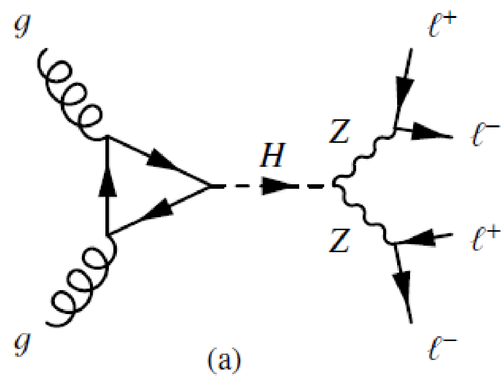


FIGURE 1.3: $\mathrm{gg} \rightarrow \mathrm{H}$ process feynman diagram.

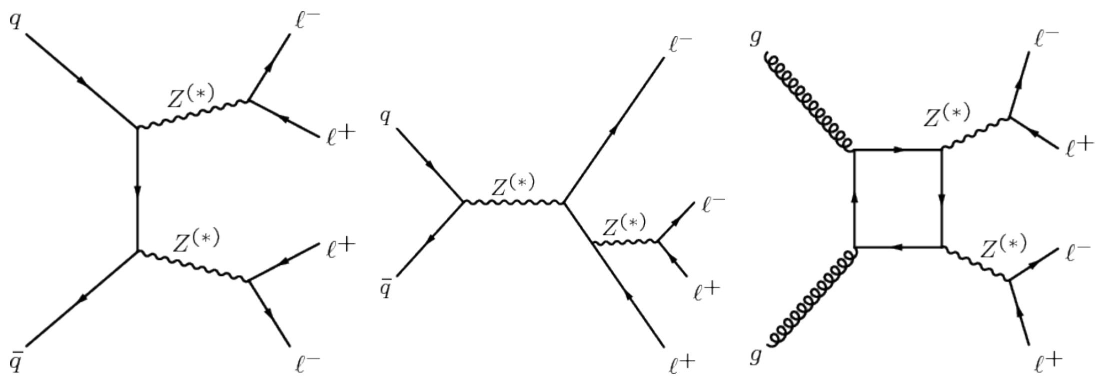
All these MC samples are produced using POWHEG for the representation of QCD effects at NLO in Higgs production, and by using JHUGEN [24, 26, 27] to represent such exotic resonances decays to four-leptons containing all the spin correlations.

Background Samples

For the estimations of the backgrounds, POWHEG V2 interfaced with PYTHIA8 was used to generate the dominant process $\mathrm{q\bar{q}} \rightarrow \mathrm{ZZ} \rightarrow 4\ell$ at the NLO accuracy level containing s-, t- and u-channel diagrams [28]. Generatars were used with same configurations as used for the Higgs signal . Feynman diagrams of the involved background processes are shown in the Figure 1.4. The dynamical renormalization and factorization scales have been chosed equal to m_{ZZ} since the simulation covers a wide range of invariant masses of ZZ.

The $\mathrm{gg} \rightarrow ZZ$ process, as shown in the Figure 1.4(c) is simulated at leading order (LO) with MCFM [29, 30]. In order to have matching in spectra of $\mathrm{gg} \rightarrow \mathrm{H} \rightarrow$

ZZ as predicted at NLO by POWHEG, the showering of these samples has been performed with some different PYTHIA8 settings. This only allows emissions only up to parton-level scale ("wimpy" shower).



(a) s-channel (b) t-channel (c) ggZZ-box

FIGURE 1.4: $\mathrm{q\bar{q}}to\mathrm{ZZ} \to 4\ell$ process s- and t- channel fynman diagrams (a) and (b). $\mathrm{gg} \to \mathrm{ZZ} \to 4\ell$ process fynman diagram (c).

The initial-state and final-state radiations are included for all events generators, which provides hints for the existence of extra (hard) photons in the event. Interference between all amplitudes of process $H \to 4\ell$ are properly taken into consideration both in POWHEG and the JHUGEN. For the generators, the used default parton distribution functions (PDFs) are NNPDF3.1 NLO (at NLO and LO) are used.

The generated events were interfaced to PYTHIA 8.209 Tune tune CUETP8M1 to include the fact of the parton shower, underlying events and hadronization.

Processing of all generated events was made via a full simulation for CMS detector using GEANT4 [31] and their reconstruction was done by exploiting same algorithms used for the data analysis. Multiple proton-proton interactions are incorporated in the simulations by reweighting the pileup distribution to match the distribution of the observed data.

Pileup Reweighting

For each year, corresponding simulation samples are reweighted to meet pileup distribution in the data. An example of the reweighting procedure for 2018 data is shown in Fig. 1.5.

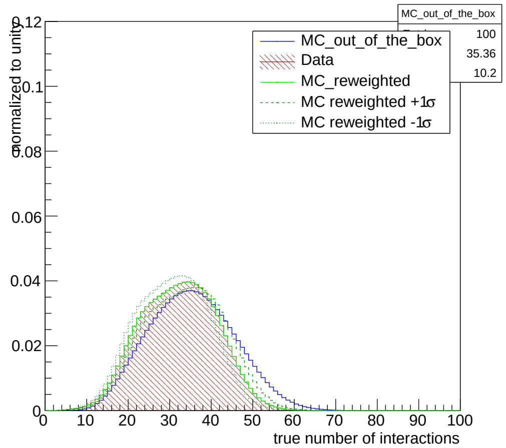


FIGURE 1.5: Pileup distribution in 2018 MC and Data shown before and after the application of PU weights. Up and down variations of 5% in the minimum bias cross section when calculating the weighs are also shown.

1.4 Analysis Strategy

As mentioned in Section 1.3, the measurements on Higgs cross sections (Inclusive and differential) are made in a fiducial region that matches the reconstruction level selection. Measurement strategy and definition of the fiducial volume is optimized to:

- detect a low mass Higgs (m_H 125.09 GeV) boson decaying to 4 leptons through a pair of Z bosons
- keep the measurements independent of how the Higgs boson is produced.

For reconstruction efficiency to be independent of fiducial phase space, it is necessary to include isolation in the definition of fiducial phase space.

In case of associated production of Higgs with vector bosons where the bosons can decay leptonically, there is possibility that Higgs is reconstructed from wrong combination of leptons. This makes it necessary to consider such "wrong combination" events as a background in order to keep complete information from unbinned simultaneous fit of signal and background parameterizations to four-lepton invariant mass.

Also, signal events which pass the fiducial level selections but fail the reconstruction are accounted in the fiducial efficiency (ϵ) and the events which pass the reconstruction but fail the fiducial selections are referred to as "nonfiducial signal" events (f_{nonfid}) and are treated as a background as illustrated in Figure 1.6 **[fid_diagram]**.

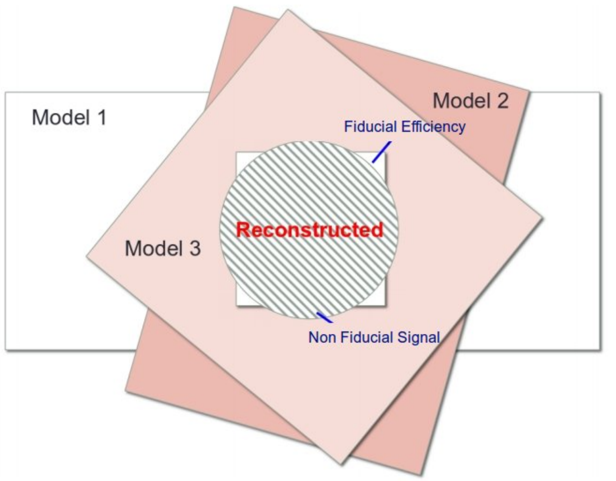


FIGURE 1.6: Fiducial efficiency(ϵ) and nonfiducial(f_{nonfid}) signal definition.

There is significant effect of imperfect resolution of the detector on the differential measurements particularly in case of jet related observables for which detector resolution is poor. A procedure, called unfolding, is applied for the correction of the reconstruction efficiencies and the effects are applied to the measurements.

The effects of imperfect detector resolution can in general have a non-negligible effect on shape of the distributions of measured observables. For this reason, for differential cross section measurements of Higgs, a procedure to correct the detection efficiencies and resolution effects is applied.
The details of the procedure would be described in next Sections.

1.4.1 Event Selection

This dissertation is based on the analysis for which event selections are similar to what was used in pervious analyses on same topic using the same final state [**CMS:2018mmw**, **CMS:2017jkd**, 32, 33]. The online event selection is based on having fired High Level Trigger paths as described in 1.2. Unlike in the analysis based on Run I dataset, we require OR of all HLT paths so that the trigger requirement does not depend on the selected final state. The reason is Full Run II data (at $137.1\ \mathrm{fb}^{-1}$), associated production modes that can come with additional leptons are targeted which improve trigger efficiency.

Vertex Selection

The events are required to have at least one primary vertex (PV) that fullfills the criteria:

- Large degree of freedom ($N_{PV} > 4$)
- Small radius of PV being ($r_{PV} < 2$ cm)
- Restricted collisions along $z-$axis ($z_{PV} < 24$ cm).

p_T threshold requirements on the Triggers

For the double lepton triggers, the requirement on minimum p_T for leading and the subleading lepton are imposed to be 17 and 8 GeV respectively; however, for triple electron triggers, it is 15, 8 and 5 GeV in the same order respectively.

Lepton Selection

Details on the selection requirments on electron and muons are explained in sections **??** and **??**, and **??** and **??** respectively. Both electrons and muons are initially required to pass POG Loose identification criteria with subsequent requirements on imparameter and isolation. These leptons are regarded as **selected leptons** and can be used for analysis.

Jet Selection

Reconstruction of the jets is mandatory to study the differential cross section measurements for the jet related observables. Jets are reconstructed with $p_T \geq 30$ GeV within $|\eta| \leq 4.7$, discussed in Section **??**. To remove overlapping of jets overlap from leptons or selected photons, it is required to have at least $\Delta R > 0.5$ where $\Delta R \equiv \sqrt{(\Delta\eta)^2 + (\Delta\phi)^2}$ [34] with respect to any of the selected leptons and FSR photons candidates.

1.4.2 ZZ Candidate Selection

Four-lepton candidates are built from the selected leptons, that pass the $\mathrm{SIP_{3D}} < 4$ vertex constraint and the isolation cuts (defined in sections **??** and **??**), where FSR photons are subtracted as briefed in Section **??**. Cross cleaning of the leptons is made by discarding electrons which are within $\Delta R < 0.05$ of selected muons.

The construction and selection of four-lepton candidates proceeds according to the following sequence:

1. **Z candidates** are built from oppositely charged and same flavoured leptons (e^+e^-, $\mu^+\mu^-$) with additional requirement on dilepton mass as $12 < m_{\ell\ell(\gamma)} < 120 GeV$. Z boson candidate is also required to include FSR photons (if any).

2. **ZZ candidates** are then built from non-overlapping Z boson candidates. The one who is reconstructed with $m_{\ell\ell}$ closer to the SM Z boson mass is regarded as $\mathrm{Z_1}$ and other as $\mathrm{Z_2}$. Additional requierements on ZZ candidates are as under:

    - **Ghost removal** : Selected four leptons are required to have $\Delta R(\eta, \phi) > 0.02$ among each other.
    - **lepton** p_T: Among selected four leptons, two are required to fulfill $p_{T,i} > 20 GeV$ and $p_{T,j} > 10 GeV$.
    - **QCD suppression**: Dilepton inivariant mass is required to pass (regardless of flavor) $m_{\ell\ell} > 4\ GeV$ for QCD induced J/Ψ veto.
    - $\mathrm{Z_1}$ **mass**: $40\ GeV < m_{\mathrm{Z1}} < 120\ GeV$
    - **four-lepton invariant mass**: $m_{4\ell} > 70\ GeV$

3. **Signal region** is built from the events containing one selected ZZ candidate at leaset.

1.4.3 Choice of best ZZ Candidate

An important change with respect to the Run I analysis is to choose the best ZZ candidate after all kinematic cuts. This helps to test other selection strategies for this candidate choice and becomes important for events having more than four selected leptons to search for Higgs boson production modes where associated particles can decay leptonically (e.g. VH and ttH).

For fiducial cross section measurement, Z_1 is chosen with mass closest to the mass of SM Z boson mass while Z_2 is chosen from Z candidates whose lepton pair gives higher p_T sum.

1.4.4 Definition of the Fiducial Volume

The measurements on Higgs cross sections (Inclusive and differential) are made in a fiducial region that matches the reconstruction level selection. This is done because fraction of all $\mathrm{H} \to 4\ell$ events which pass the selection criteria as described in Section 1.4.3 can significantly differ for different Higgs production mechanisms and also for its distinct exotic models. As a treatment for the impact of model dependent acceptances, these measurements are done in the a certain fiducial volume which is defined so as to closely match with the experimental acceptance from reconstruction level selections.

Requirement of fiducial phase space is minimum four leptons having $p_T > 7$ GeV and $|\eta| < 2.5$ in case of electron, and $p_T > 5$ GeV and $|\eta| < 2.4$ in case of muon. Besides, the leading and sub-leaing leptons are required to have $p_T > 20$ GeV and $p_T > 10$ GeV respectively. The four-momenta of each lepton is calculated at the hard scattering level and isolation is calculated by adding p_T of all stable particles within a cone size $\Delta R < 0.3$ around the lepton (by subtracting contributions from neutrinos and FSR photons). All neutrinos and FSR photons are excluded from above summation. Then the lepton's relative isolation, defined by ratio of this sum and p_T of considered lepton, is required to be less than 0.35 being consistent with reconstruction level isolation requirement [32].

Simulation based studies show that signal efficiency can significantly change among different models without requiring lepton isolation at fiducial level. The difference is more visible in $t\bar{t}\mathrm{H}$ and $\mathrm{gg} \to \mathrm{H}$ production modes as can be seen in the Table **??**. It is understood from the fact that $t\bar{t}\mathrm{H}$ events have more hadronic activity around leptons than those of the $\mathrm{gg} \to \mathrm{H}$ events, so the cut at relative isolation removes more events in $t\bar{t}\mathrm{H}$ than $\mathrm{gg} \to \mathrm{H}$. Hence, for reconstruction efficiency to be independent of fiducial phase space, it is necessary to include isolation in the definition of fiducial phase space.

For other requirements of fiducial volume, there must be two pairs of same falvor oposite sign (SFOS) leptons where Z_1 being the one with invariant mass closest to the SM Z boson and reconstructed with $40 \leq m(\mathrm{Z}_1) \leq 120$ GeV. Z_2 is reconstructed from rest of the SFOS pairs of leptons within invariant mass range of $12 \leq m(\mathrm{Z}_2) \leq 120$ GeV. In case of more than one Z_2 passing the criteria, leptons pair with highest $\Sigma|p_T|$ is selected. To remove the overlapping between leptons, we ask for $\Delta R(\ell_i \ell_j) > 0.02$ for any $i \neq j$. Any opposite-charge pair of leptons is required to meet the condition $m(\ell^+\ell'^-) \geq 4$ GeV to suppress background from QCD for low mass resonance which is called J/ψ veto.

Invariant mass of selected four-leptons is required to fulfill 105 GeV $< \mathrm{m}(4\ell) <$ 140 GeV and must originate from Higgs boson decay. This helps to avoid from the contamination from off-shell gg fusion production [35, 36]. The requirements that define fiducial phase space are also gathered in Table 1.1.

TABLE 1.1: Overview of all requirements utilized to define the fiducial phase space.

Requirements on $\mathrm{H} \to 4\ell$ **fiducial phase space**	
Lepton kinematics and the isolation	
p_T of the leading lepton	$p_T > 20$ GeV
p_T of sub-leading lepton	$p_T > 10$ GeV
Additional electrons (muons) p_T	$p_T > 7$ (5) GeV
Pseudorapidity of electrons (muons)	$\lvert\eta\rvert < 2.5$ (2.4)
Sum of the scalar p_T of all the stable objects within $\Delta R < 0.3$ from the lepton	$< 0.35 p_T$
Event topology	
From leptons satisfying criteria above, minimum two SFOS lepton pairs	
Invarivant mass of Z_1 candidate	$40 < m(\mathrm{Z}_1) < 120$ GeV
Invariant mass of Z_2 candidate	$12 < m(\mathrm{Z}_2) < 120$ GeV
Separation between selected four leptons	$\Delta R(\ell_i \ell_j) > 0.02$
Invariant mass of any OS lepton pair	$m(\ell_i^+ \ell_j^-) > 4$ GeV
Invariant mass of selected four leptons	$105 < m_{4\ell} < 140$ GeV

Part of the signal events passing fiducial phase space definition ($\mathcal{A}_{\text{fid}}$), and reconstruction efficiencies (ϵ) for SM production modes are registered in Table 1.2.

1.4.5 Signal Modeling

For fiducial cross section measurements, we perform a maximum likelihood fit of the signal and background parameterisations to the observed $m_{4\ell}$ mass distribution and directly extract the parameter of interest (σ_{fid}). Definition of the likelihood function and the procedrue is explained in Section 1.4.7.

TABLE 1.2: Summary of different SM signal models. The MC samples are from full Run 2 yearly productions (2016, 2017 and 2018) assuming $m_{\mathrm{H}} = 125.09$ GeV. Luminosity averaged fiducial acceptance ($\mathcal{A}_{\mathrm{fid}}$) defined the fraction of events passing fiducial phase space selections, reconstruction efficiency (ϵ) represented by the ratio of reconstructed to accepted events. Also, f_{nonfid} which are reconstructed events but does not pass fiducial phase space definition at generator level. Values are obtained using 13 TeV signal models and the uncertainties shown are only statistical uncertainties.

Signal process	$\mathcal{A}_{\mathrm{fid}}$	ϵ	f_{nonfid}	$(1+f_{\mathrm{nonfid}})\epsilon$
Individual production modes of Higgs boson				
gg→H (POWHEG)	0.402 ± 0.001	0.592 ± 0.002	0.053 ± 0.001	0.624 ± 0.002
VBF (POWHEG)	0.444 ± 0.002	0.605 ± 0.003	0.043 ± 0.001	0.631 ± 0.003
WH (POWHEG+MINLO)	0.325 ± 0.002	0.588 ± 0.003	0.075 ± 0.002	0.632 ± 0.004
ZH (POWHEG+MINLO)	0.340 ± 0.003	0.594 ± 0.005	0.081 ± 0.004	0.643 ± 0.006
ttH (POWHEG)	0.314 ± 0.003	0.585 ± 0.006	0.169 ± 0.006	0.684 ± 0.007

However, in this section we describe the PDF used to describe the signal shapes for different masses of Higgs. The resolution function model depends only on the leptons quality, while the exact shape depends on the lepton kinematics, and so depend also on the Higgs mass. Double Crystal Ball function is used for shape description of signal resonant contribution ($\mathcal{P}_{\mathrm{res}}(m_{4\ell})$) [32] with the yield proportional to the parameter of interest (σ_{fid}). A Double Crystal-Ball function $f_{dCB}(m_{4\ell}\,|\,m_H)$ is defined as under:

$$dCB(\xi) = N \cdot \begin{cases} A \cdot (B + |\xi|)^{-n_L}, & \text{for } \xi < \alpha_L \\ A \cdot (B + |\xi|)^{-n_R}, & \text{for } \xi > \alpha_R \\ \exp\left(-\xi^2/2\right), & \text{for } \alpha_L \leq \xi \leq \alpha_R \end{cases} \tag{1.1}$$

where $\xi = (m_{4\ell} - m_H - \Delta m_H)/\sigma_m$.

This function has six independent parameters, and is intended to capture the Gaussian core (σ_m) of the four-lepton mass resolution function, systematic mass shift Δm_H of the peak, and the left- and right-hand tail originating from leptons emitting brem in the tracker material, present for both electrons and muons, and from the non-Gaussian mis-measurements specific to interactions of electrons with the detector material (two parameters, n and α, for each side of the mean): The prominence of the left-,right-hand tail is defined the power n_L, n_R, respectively. The parameters α_L, α_R define where the splicing of the tails and the core are made, in units of σ_m. Parameters A and B are not independent; they are defined by requiring the continuity of the function itself and its first derivatives. N is the normalizing constant.

To modell the Higgs shape, several mass points in mass points 120-130 GeV are used in the three final states. The fitting strategy used to deal with this situation is to use the linear approximation of all the Double Crystal Ball parameters varying with with m_H.

$$params = parms_{CB0} + params_{CB1} \times (m_H - 125GeV) + params_{CB2} \times (m_H - 125GeV)^2 \quad (1.2)$$

Simultaneous fits are performed using 5 mass samples to extract the $parms_{CB0}$, $params_{CB1}$ and $params_{CB2}$. The initial value for the $parms_{CB0}$ is obtained by fitting 125 GeV mass sample alone, and refitted in the simultaneous fits. Uncertainties on these parameters are included as systematic uncertainties on the measurements as described in the Section **??**.

Some examples of the fits are shown in Fig. 1.7 for the ggH production mode. The simultaneous fit at different mass points are shown in Fig. 1.8.

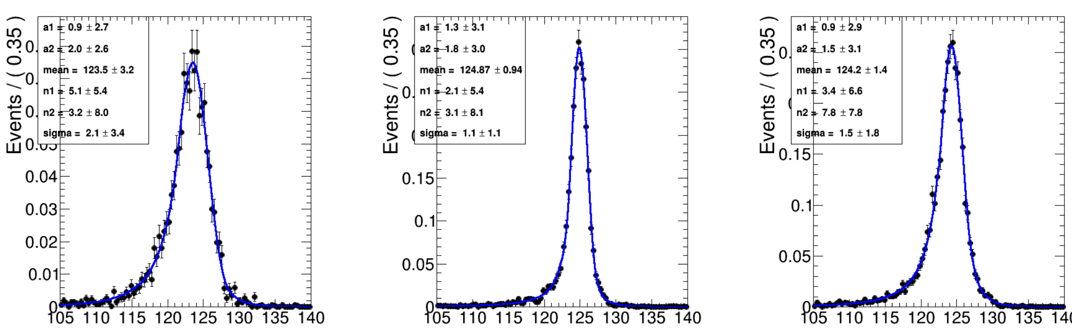


FIGURE 1.7: Probability density functions $f(m_{4l}|m_H)$ for signal with m_H=**125** GeV at the reconstruction level after the full lepton and event selections are applied. The distributions obtained **from 13** TeV ggH MC samples are fitted with the model described in the text for 4μ (left), $4e$ (center) and $2e2\mu$ (right) events. The distribution show the events for p_T below 10 GeV.

1.4.6 Background Modeling

Background processes for this analysis can be divided into two parts:

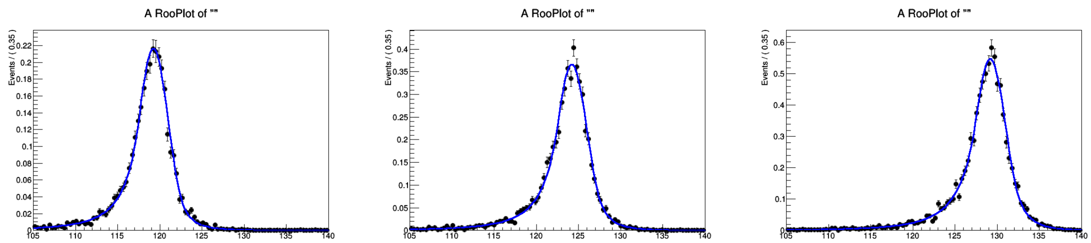


FIGURE 1.8: Simultaneous fits for $f(m_{4l}|m_H)$ for $\mathrm{gg} \rightarrow \mathrm{H}$ signal for $p_T$below 10 GeV with m_H**=120,125,130** GeV at the reconstruction level after the full lepton and event selections are applied. The distributions obtained **from 13** TeV $\mathrm{gg} \rightarrow \mathrm{H}$ MC samples are fitted with the model described in the text for $2e2\mu$ events.

1. **Irreducible Background**: Mainly it is due to quark anti-quark ($\mathrm{q\bar{q}}$) annihilation which produce a pair of Z bosons and each Z boson decays into a pair of leptons resulting four leptons in the final state. The second one is when quark anti-quark ($\mathrm{q\bar{q}}$) annihilate to Z boson which further decays into a pair of leptons. One of the two leptons, then radiate a virtual Z boson which subsequently decays into a pair of leptons. This background has dominant contribution in the analysis. Additional source of the Irreducible background is from gluon gluon fusion process. However, in the analysis it has sub-dominant contribution. Feynman diagrams of the these background processes can be seen in the diagram **??**.

2. **Reducible Background**: This background is instrumental nature and arises from the process where partial jets are misidentified as leptons. The main processes which contributes in this background are: production of jets in association with Z boson, production of the jets in association with WZ boson pair and the $\mathrm{t\bar{t}}$ pair production. This background is denoted as Z+X.

For fiducial cross section measurement for $\mathrm{H} \rightarrow 4\ell$, both of irreducible backgrounds ($\mathrm{q\bar{q}} \rightarrow \mathrm{ZZ}$ and gg $\rightarrow$ ZZ) are estimated from Monte Carlo (MC) simulation as done earlier in Ref. [32].

$\mathrm{q\bar{q}} \rightarrow \mathrm{ZZ}$ **Modelling**

The $\mathrm{q\bar{q}} \rightarrow \mathrm{ZZ}$ background is generated at NLO, while the fully differential cross section has been computed at NNLO [**Grazzini2015407**]. Since the cross sections are not yet available in the generator we use for its estimation, the NNLO/NLO

k-factors for the $q\bar{q} \to ZZ$ background process are applied to the POWHEG sample used. The inclusive cross sections obtained using the same PDF and renormalization and factorization scales as the POWHEG sample at LO, NLO, and NNLO are shown in Table 1.3. The NNLO/NLO k-factors are applied in the analysis differentally as a function of $m(ZZ)$.

Additional NLO electroweak corrections which depend on the initial state quark flavor and kinematics are also applied to the $q\bar{q} \to ZZ$ background process in the region $m(\mathrm{ZZ}) > 2m(\mathrm{Z})$ where the corrections have been computed. The differential QCD and electroweak k-factors can be seen in Figure 1.9.

QCD Order	$\sigma_{2\ell2\ell'}(\mathrm{fb})$	$\sigma_{4\ell}(\mathrm{fb})$
LO	$218.5^{+16\%}_{-15\%}$	$98.4^{+13\%}_{-13\%}$
NLO	$290.7^{+5\%}_{-8\%}$	$129.5^{+4\%}_{-6\%}$
NNLO	$324.0^{+2\%}_{-3\%}$	$141.2^{+2\%}_{-2\%}$

TABLE 1.3: Cross sections for $q\bar{q} \to ZZ$ production at 13 TeV

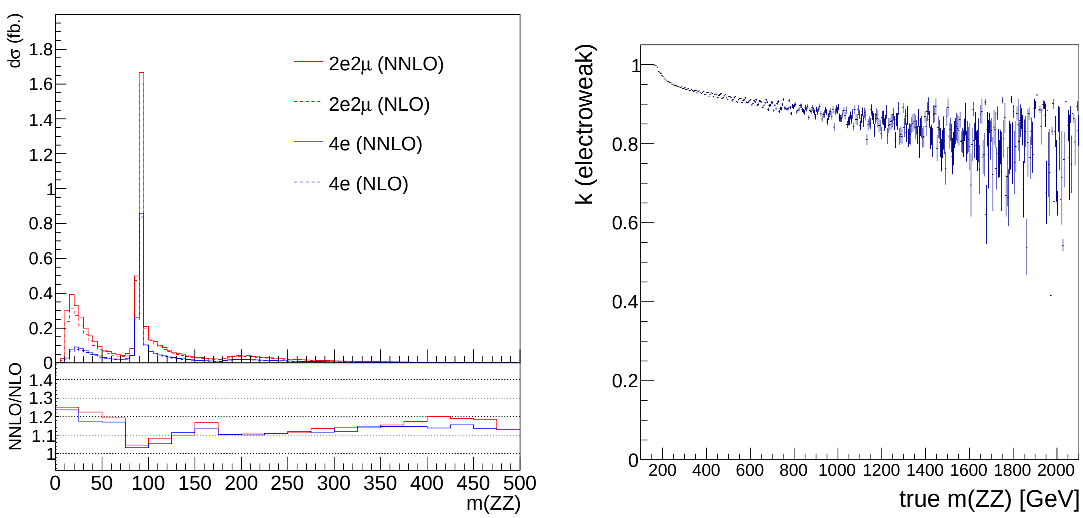


FIGURE 1.9: Left: NNLO/NLO QCD k-factor for the $q\bar{q} \to ZZ$ background as a function of $m(ZZ)$ for the 4ℓ and $2\ell2\ell'$ final states. Right: NLO/NLO electroweak k-factor for the $q\bar{q} \to ZZ$ background as a function of $m(ZZ)$.

gg → ZZ Modelling

Event simulation for the gg → ZZ background is done at LO with the generator 7.0 [**Campbell:2011bn**, 29, 30]. Although no exact calculation exists beyond the LO for the gg → ZZ background, it has been recently shown [**Bonvini:1304.3053**] that the soft collinear approximation is able to describe the background cross section and the interference term at NNLO. Further calculations also show that the K factors are very similar at NLO for the signal and background [**Melnikov:2015laa**] and at NNLO for the signal and interference terms [**Li:2015jva**]. Therefore, the same K factor is used for the signal and background [**Passarino:1312.2397v1**]. The NNLO K factor for the signal is obtained as a function of $m_{4\ell}$ using the HNNLO v2 Monte Carlo program [**Catani:2007vq**, **Grazzini:2008tf**, **Grazzini:2013mca**] by calculating the NNLO and LO $\mathrm{gg} \to H \to 2\ell 2\ell'$ cross sections at the small H boson decay width of 4.07 and taking their ratios. The NNLO as well as the NLO K factors and the cross sections from which they are derived are illustrated in Fig. 1.10, along with the NNLO, NLO and LO cross sections at the SM H boson decay width [**Heinemeyer:2013tqa**].

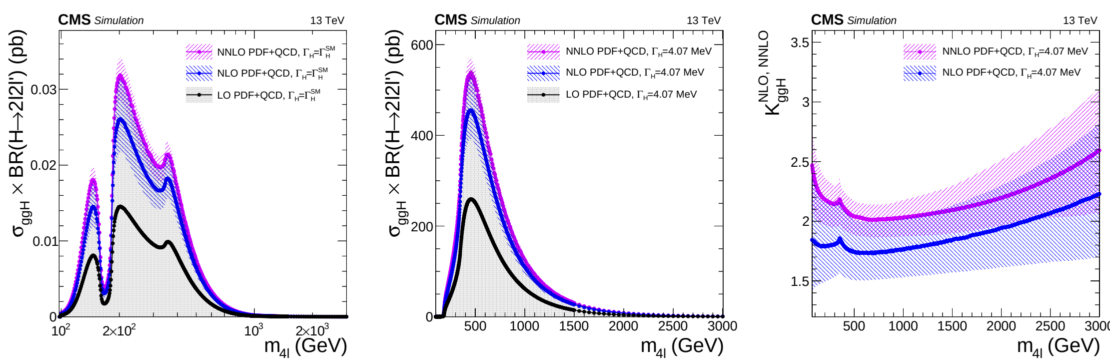


FIGURE 1.10: $\mathrm{gg} \to H \to 2\ell 2\ell'$ cross sections at NNLO, NLO and LO at each H boson pole mass using the SM H boson decay width (top) or at the fixed and small decay width of 4.07 MeV (top). The cross sections using the fixed value are used to obtain the K factor for both the signal and the continuum background contributions as a function of $m_{4\ell}$ (bottom).

In addition, following the study in Ref. [**Bonvini:2013**], in these measurements the expected yield for the gg → ZZ background has been increased by assigning the same ($m_{4\ell}$ dependent) k-factor as signal $\mathrm{H} \to 4\ell$ process to leading order (LO) cross section of background. In the region ($105 < m_{4\ell} < 140$ GeV) this factor leads to an increase of expected gg → ZZ background by factor of ~ 2.66. After inclusion of this k-factor, the gg $\to ZZ$ process is still a sub-dominant background in all measurements.

Reducible Background Estimation

In the $H \to ZZ \to 4\ell$ analysis, Reducible background (Z+X) is estimated using data with tight-to-loose "fake rate" technique is used in the control regions of data by following Ref. [32].

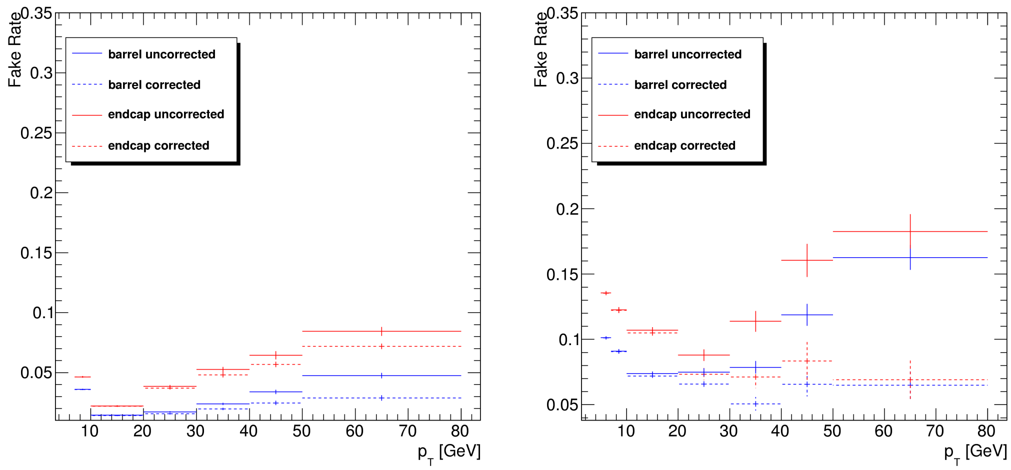
In the $H \to ZZ \to 4\ell$ analysis, Reducible Background (Z+X) is estimated using data with tight-to-loose "fake rate (FR)" technique. The rates of involved background processes are estimated by measuring the probabilities for fake electrons and fake muons (denoted by f_e and f_μ respectively) which do pass the **loose** selection criteria (defined in Section **??** and **??**) to also pass the final selection criteria (defined in Section 1.4.2) [32]. These probabilitiess or fake rates (FR) are applied in dedicated control samples in order to extract the expected background yield in the signal region.

Two independent methods are presented to measure both the yields and shapes of the reducible background. The final result combines the outcome of the two approaches.

(1) Opposite Sign (OS) method Fore lepton fake ratios f_e, f_μ, samples of $Z(\ell\ell)+e$ and $Z(\ell\ell)+\mu$ events are selected which are expected to be completely dominated by final states which include a Z boson and a fake lepton. These events are required to have two SFOS leptons with $p_T > 20/10$ GeV (p_T of leading and subleading leptons) passing the tight selection criteria, thus forming the Z candidate. In addition, there is exactly one lepton passing the loose selection criteria as defined above. This lepton is used as the probe lepton for the fake ratio measurement. The invariant mass of this lepton and the opposite sign lepton from the reconstructed Z candidate should satisfy $m_{2l} > 4$ GeV.

To reduce the contribution from photon (asymmetric) conversions populating low masses, FRs are evaluated using the tight requirement $|M_{inv}(\ell_1, \ell_2) - M_Z| <$ 7 GeV. The fake ratios are measured in bins of the p_T loose lepton in barrel and endcap region.

The electron and muon fake rates are measured within $|M_{inv}(\ell_1, \ell_2) - M_Z| < 7$ GeV and $E_{\mathrm{T}}^{\mathrm{miss}} < 25$ GeV are shown in Figure 1.11.



(a) (b)

FIGURE 1.11: Fake rates as a function of the probe p_T for electrons (a) and muons (b) which satisfy the loose selection criteria, measured in a $Z(\ell\ell) + \ell$ sample in the 13 TeV data. The barrel selection includes electrons (muons) up to $|\eta|$ = 1.479 (1.2). The Fake rates are shown before (dotted lines) and after (plain line) removal of WZ contribution from MC.

Fake Rate Application (OS Method) Two control samples are obtained as subsets of four lepton events which pass the first step of the selection (*First* Z step, see section 1.4.2), requiring an additional pair of loose leptons of same flavour and opposite charge, that pass the $\mathrm{SIP_{3D}}$ cut. The events must satisfy all kinematic cuts applied for the *Higgs phase space* selection (see 1.4.2). The first control sample is obtained by requiring that the two loose leptons which do not make the Z_1 candidate do not pass the final identification and isolation criteria. The other two leptons pass the final selection criteria by definition of the Z_1. This sample is denoted as "2 Prompt + 2 Fail" (*2P+2F*) sample. It is expected to be populated with events that intrinsically have only two prompt leptons (mostly DY, with a small fraction of $t\bar{t}$ and $Z\gamma$ events). The second control sample is obtained by requiring one of the four leptons not to pass the final identification and isolation criteria. The other three leptons should pass the final selection criteria. This control sample is denoted as "3 Prompt + 1 Fail" (*3P+1F*) sample. It is expected to be populated with the type of events that populate the 2P+2F region, although with different relative proportions, as well as with WZ events that intrinsically have three prompt leptons.

The control samples obtained in this way, orthogonal by construction to the signal region, are enriched with fake leptons and are used to estimate the reducible background in the signal region.

The invariant mass distribution of events selected in the 2P+2F control sample is shown in Fig. 1.12 for the 13 TeV dataset.

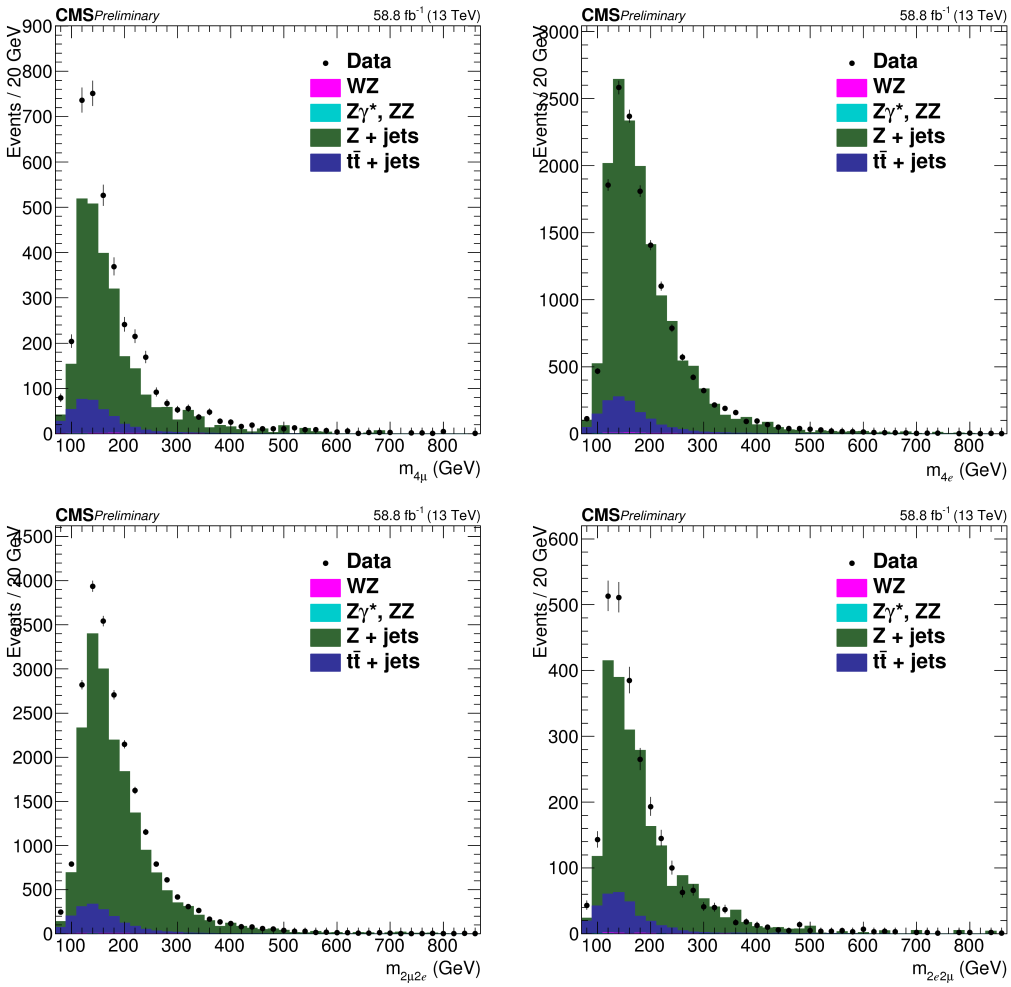


FIGURE 1.12: Invariant mass distribution of the events selected in the 2P+2F control sample in the 13 TeV dataset, (top left) 4μ , (top right) $4e$, (bottom left) $2\mu 2e$ and (bottom right) $2e2\mu$ channels.

The expected number of reducible background events in the 3P+1F region, $N^{\text{bkg}}_{\text{3P1F}}$, can be computed from the number of events observed in the 2P+2F control region, N_{2P2F}, by weighting each event in the region with the factor $(\frac{f_i}{1-f_i} + \frac{f_j}{1-f_j})$, where f_i and f_j correspond to the fake ratios of the two loose leptons:

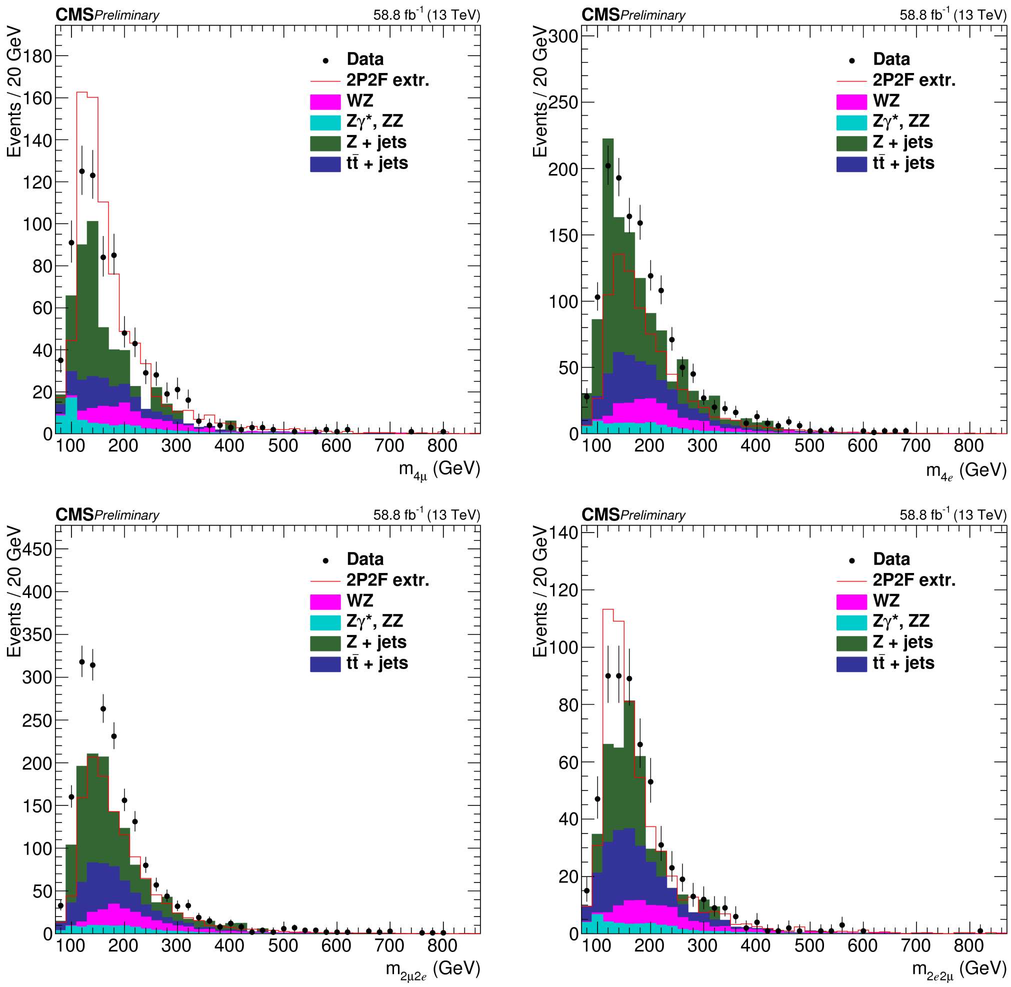


FIGURE 1.13: Invariant mass distribution of the events selected in the 3P+1F control sample in the 13 TeV dataset, (top left) 4μ , (top right) $4e$, (bottom left) $2\mu 2e$ and (bottom right) $2e2\mu$ channels.

$$N^{\text{bkg}}_{\text{3P1F}} = \sum \left(\frac{f_i}{1-f_i} + \frac{f_j}{1-f_j}\right) N_{\text{2P2F}} \tag{1.3}$$

Figure 1.13 shows the invariant mass distributions of the events selected in the 3P+1F control sample, together with the expected reducible background estimated from Eq. 1.3, stacked on the distribution of WZ and of irreducible background ($ZZ, Z\gamma* \to 4\ell$) taken from the simulation.

Would the fake rates be measured in a sample that has exactly the same background composition as the 2P+2F sample, the difference between the observed number of events in the 3P+1F sample and the expected background predicted from the 2P+2F sample would solely amount to the (small) WZ and $Z\gamma_{conv}$ contribution. Large differences arise because the fake rates used in Eq. 1.3 do not properly account for the background composition of the 2P+2F control sample.

In particular, the difference seen in Fig. 1.13 between the observed 3P+1F distribution and the expectation from 2P+2F, in the channels with loose electrons ($4e$ and $2\mu 2e$), and concentrated at low masses, is due to photon conversions. This is confirmed explicitly by the simulation. The difference between the 3P+1F observation and the prediction from 2P+2F to recover the missing contribution from conversions - and more generally, in principle, to "correct" for the fact that the fake rates do not properly account for the background composition of the 2P+2F sample. More precisely, the expected reducible background in the signal region is given by the sum of two terms :

- a "2P2F component", obtained from the number of events observed in the 2P+2F control region, N_{2P2F}, by weighting each event in that region with the factor $\frac{f_i}{1-f_i}\frac{f_j}{1-f_j}$, where f_i and f_j correspond to the fake ratios of the two loose leptons.

- a "3P1F component", obtained from the difference between the number of observed events in the 3P+1F control region, N_{3P1F}, and the expected contribution from the 2P+2F region and ZZ processes in the signal region, $N^{\mathrm{ZZ}}_{\mathrm{3P1F}} + N^{\mathrm{bkg}}_{\mathrm{3P1F}}$. The $N^{\mathrm{bkg}}_{\mathrm{3P1F}}$ is given by equation 1.3 and $N^{\mathrm{ZZ}}_{\mathrm{3P1F}}$ is the contribution from ZZ which is taken from the simulation. The difference $N_{\mathrm{3P1F}} - N^{\mathrm{bkg}}_{\mathrm{3P1F}} - N^{\mathrm{ZZ}}_{\mathrm{3P1F}}$, which may be negative, is obtained for each (p_T, η) bin of the "F" lepton, and is weighted by $\frac{f_i}{1-f_i}$, where f_i denotes the fake rate of this lepton. This "3P1F component" accounts for the contribution of reducible background processes with only one fake lepton (like WZ events), and for the contribution of other processes (e.g. photon conversions) that are not properly estimated by the 2P2F component, because of the fake rates used.

Therefore, the full expression for the prediction can be symbolically written as:

$$N^{bkg}_{\mathrm{SR}} = \sum \frac{f_i}{(1-f_i)}(N_{\mathrm{3P1F}} - N^{\mathrm{bkg}}_{\mathrm{3P1F}} - N^{\mathrm{ZZ}}_{\mathrm{3P1F}}) + \sum \frac{f_i}{(1-f_i)}\frac{f_j}{(1-f_j)}N_{\mathrm{2P2F}} \qquad (1.4)$$

Previous equation is equivalent to the following:

$$N_{\mathrm{SR}}^{bkg} = (1 - \frac{N_{3P1F}^{ZZ}}{N_{3P1F}}) \sum_{j}^{N_{3P1F}} \frac{f_a^j}{1-f_a^j} - \sum_{i}^{N_{2P2F}} \frac{f_3^i}{1-f_3^i} \frac{f_4^i}{1-f_4^i} \tag{1.5}$$

For channels where the Z_2 candidate is made from two electrons, the contribution of the 3P1F component is positive, and amounts to typically 30% of the total predicted background.

For channels with loose muons (4μ and $2e2\mu$), the 3P+1F sample is rather well described by the prediction from 2P+2F, as seen in Fig. 1.13b, and the 3P1F component is mainly driven by statistical fluctuations in the 3P+1F sample, which are larger than the expectation from WZ production.

Table 1.4 shows the expected number of events in the signal regions from the reducible background processes for the 13 TeV.

Channel	4e	4μ	$2e2\mu$	$2\mu2e$
2016	20.2 ± 6.2	27.0 ± 8.58	25.3 ± 8.0	22.3 ± 6.8
2017	16.1 ± 4.9	33.1 ± 10.4	24.3 ± 7.8	21.3 ± 6.5
2018	25.1 ± 7.6	50.7 ± 15.7	35.0 ± 11.0	32.7 ± 9.9

TABLE 1.4: The contribution of reducible background processes in the signal region predicted from measurements in 2016, 2017 and 2018 data using the OS method. The predictions correspond to 35.9, 41.5 and 59.7 fb^{-1} of data at 13 TeV, respectively.

(2) Same Sign (SS) method This method used to predict the reducible background allows to have an inclusive measurement of the all the main reducible backgrounds at the same time.

The control sample is obtained as a subset of the events that satisfy the first step of the selection (*First Z* step, see section 1.4.2), requiring an additional pair of loose leptons of same sign (to avoid signal contamination) and same flavour (SS-SF: $e^{\pm}e^{\pm}$, $\mu^{\pm}\mu^{\pm}$) . The SS-SF leptons are requested to pass the $\mathrm{SIP_{3D}}$ cut while no identification or isolation requirements are imposed. The reconstructed invariant mass of the SS-SF leptons has to satisfy $m_{\ell\ell} > 12$ GeV, $m_{\ell\ell} < 120$ GeV, the reconstructed four-lepton invariant mass is required to satisfy $m_{4\ell} > 100$, and the QCD suppression cut (see 1.4.2) is applied. The inclusive number of reducible background events in the signal region is derived from this set of events and from the probability for the two additional leptons to pass the isolation and identification analysis cuts, obtained from the fake rate measurement presented in section 1.4.6.

Starting from the control sample previously described, the final reducible background prediction in the signal region is given by the following expression:

$$N^{\mathrm{Z+X}}_{\mathrm{expect}} = \quad N^{\mathrm{DATA}} \times (\tfrac{\mathrm{OS}}{\mathrm{SS}})^{\mathrm{MC}} \times \quad f_1 \times f_2 \tag{1.6}$$

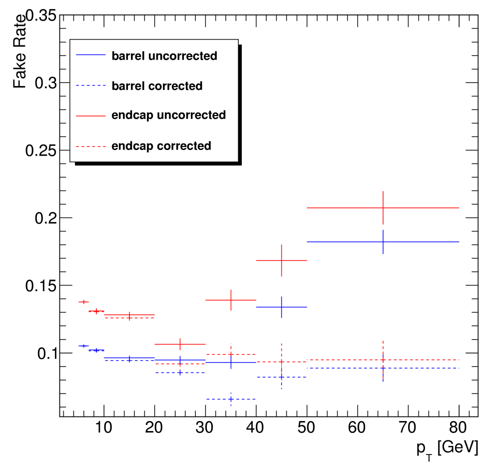
where:

- N^{DATA} is the number of events in the control region,
- $(\frac{\mathrm{OS}}{\mathrm{SS}})^{\mathrm{MC}}$ is a correction factor between opposite sign and same sign control samples,
- f_1 and f_2 are the fake rates of each additional loose lepton, parameterised as a function of p_T and η.

Fake Rate Determination (SS Method) The lepton fake rates are determined in the very similar way as it was described before for the Opposite Sign method. Samples of $Z(\ell\ell) + e$ and $Z(\ell\ell) + \mu$ events are selected in the same way except that the mass window around the nominal Z mass is set to 40 GeV $< M_{inv}(\ell_1, \ell_2) <$ 120 GeV, as in the signal region ("SS phase space"). Also, despite the cut on the missing transverse energy, a contamination of real leptons from WZ events is still visible at high p_T. This contribution is thus subtracted from the final fake rate, using the estimation given by the Monte Carlo. The fake rates for muon are shown in Fig. 1.14 for both muons in the barrel ($|\eta| < 1.2$) and in the endcaps ($|\eta| > 1.2$).

For electrons, events where a radiated photon makes an asymmetric conversion, where one low p_T leg is not identified, contribute to the $Z + e$ sample that is used to measure the electron fake rate. While the requirement $|M_{inv}(\ell_1, \ell_2) - M_Z| <$ 7 GeV ("OS phase space") largely suppresses FSR of photons radiated off the lepton legs, these radiations occur at a much larger rate in the "SS phase space". As a result of this enhanced contribution from conversions, the electron fake rates measured within the SR phase space are larger than the "OS fake rates".

However, the relative fraction of FSR conversions is not the same in the "SS phase space" and in the control sample (defined below) where the fake rates will be applied. A correction accounting for this difference must be applied to the fake rates measured within the SS phase space, in order to obtain average fake rates that



(a)

FIGURE 1.14: Fake rates as a function of the probe p_T for muons which satisfy the loose selection criteria, measured in a $Z(\ell\ell) + \ell$ sample in the 13 TeV data. The barrel selection includes electrons (muons) up to $|\eta|$ = 1.479 (1.2). The Fake rates are shown before (dotted lines) and after (plain line) the reoval of the WZ contribution from the simulation.

are appropriate for the control sample. To determine this correction, several fake rate samples of $Z(ll)+e$ events are defined by applying different cuts on $M_{inv}(\ell_1, \ell_2)$ or on $M_{inv}(\ell_1, \ell_2, e)$. In addition to the "OS fake rate sample", where $|M_{inv}(\ell_1, \ell_2) - M_Z| < 7$ GeV ensures a minimal amount of conversions from FSR photons, one defines a sample that is maximally enriched in FSR conversions by $|M_{inv}(\ell_1, \ell_2, e) - M_Z| < 5$ GeV, as well as samples with an intermediate contamination from FSR conversions (60 GeV $< M_{inv}(\ell_1, \ell_2) <$ 120 GeV). In each sample, one determines, in several (p_T, η) bins for the loose electron:

- the fake rate ratio;
- the average value of the expected inner missing hits ($N_{miss.hits}$ in the following), variable useful to tag conversions.

In a given (p_T, η) bin, both the measured fake rate and the average $< N_{miss.hits} >$ are expected to grow linearly with the fraction of conversions. Hence, one expects a linear dependence of the fake rate with respect to $< N_{miss.hits} >$. This linear behaviour is demonstrated in Fig. 1.15, and linear fits are made in each (p_T, η) bin, which relate the fake rate to $< N_{miss.hits} >$.

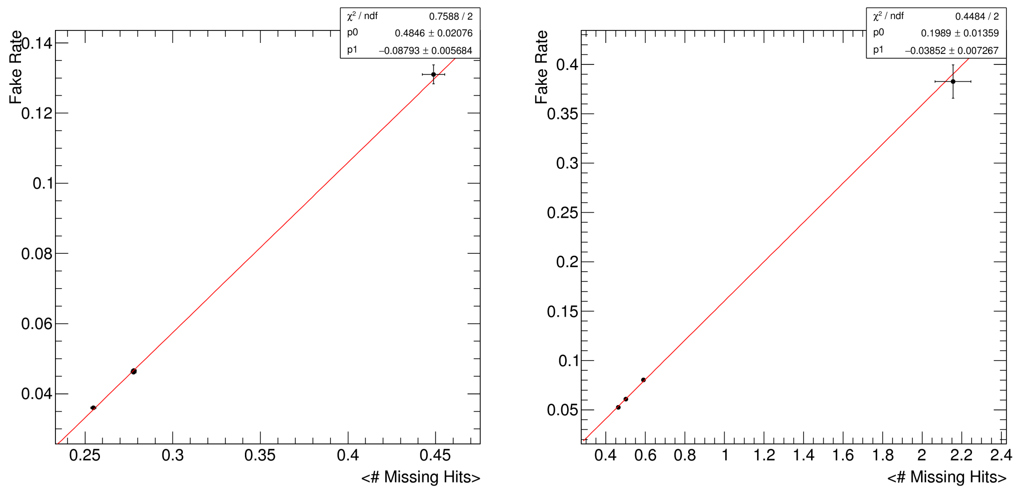
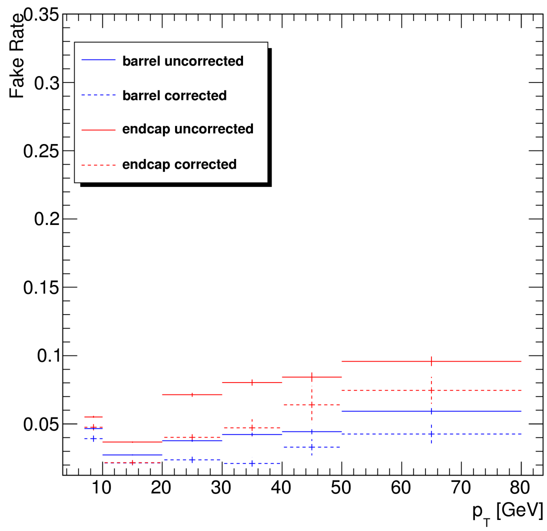


FIGURE 1.15: Examples of the correlation between the fake rate and the fraction of loose electrons for which the track has one missing hit in the pixel detector for $7 < p_T < 10$ GeV electrons in the barrel (left) and for $20 < p_T < 30$ GeV electrons in the endcaps (right). Each dot shows the measurements made in a given Z + loose e sample. Results are produced at 13 TeV with 36.8 fb^{-1} data.

Finally, one looks at the loose electrons in the control sample where the fake rate will be applied, and one measures, in each (p_T, η) bin, $< N_{miss.hits} >$. The proper average fake rate corresponding to the control sample is then determined from the linear relation derived previously. Figure 1.16 shows examples of the resulting corrected fake rates, together with the fake rates measured in the SS phase space. It can be seen that the correction for the actual fraction of conversions that is present in the control sample lowers the fake rates considerably with respect to what is measured in the SS phase space. The determination of these corrected fake rates mostly suffers from the limited statistics of the control sample, which translates into a large uncertainty on $< N_{miss.hits} >$, the error on each dot being the fake rate obtained from the linear relations using the error on the $< N_{miss.hits} >$.

Fake Rate Application (SS Method) Figure 1.17 shows the invariant mass distribution of events in control samples with SS-SF loose leptons, for channels with loose electrons ($4e$ and $2\mu 2e$) and channels with loose muons (4μ and $2e2\mu$), for the 13 TeV dataset. The prediction from the Monte-Carlo simulation is also shown.

A good agreement is achieved between data and MC for the channels with loose electrons. The agreement is not as good for loose muons but this has no



(a)

FIGURE 1.16: Average fake rates to be applied to the control sample of SS Method (plain line), compared to the fake rates measured in the SS phase space (dotted line), for electrons in the barrel (blue) and in the endcaps (red). Results are produced at 13 TeV with 2018 data.

impact on the final result as only data are used in the end.

The differences in rates between OS and SS samples are used to compute the correction factor in equation 1.6 for the final data-driven estimation. They are given in table 1.5.

Channel $\rightarrow$	4e	$2\mu 2$e	4μ	$2e2\mu$
2016	1.00 ± 0.01	1.00 ± 0.03	1.03 ± 0.03	1.00 ± 0.01
2017	1.01 ± 0.01	1.04 ± 0.03	1.01 ± 0.03	1.00 ± 0.01
2018	1.01 ± 0.01	1.00 ± 0.01	1.03 ± 0.02	1.03 ± 0.03

TABLE 1.5: The OS/SS ratios used to estimate the number of ZX events with the SS method for each final state in all three years.

Combination of the Z+X background estimtaions Two methods for the reducible background estimation are presented with associated uncertainties in the previous sections. Table 1.6 shows the summary of the results of the methods. The shape systematics is absorbed in the statistical and systematic uncertainties on the estimated yields, and the total uncertainties are found to be of the order of $35-45\%$.

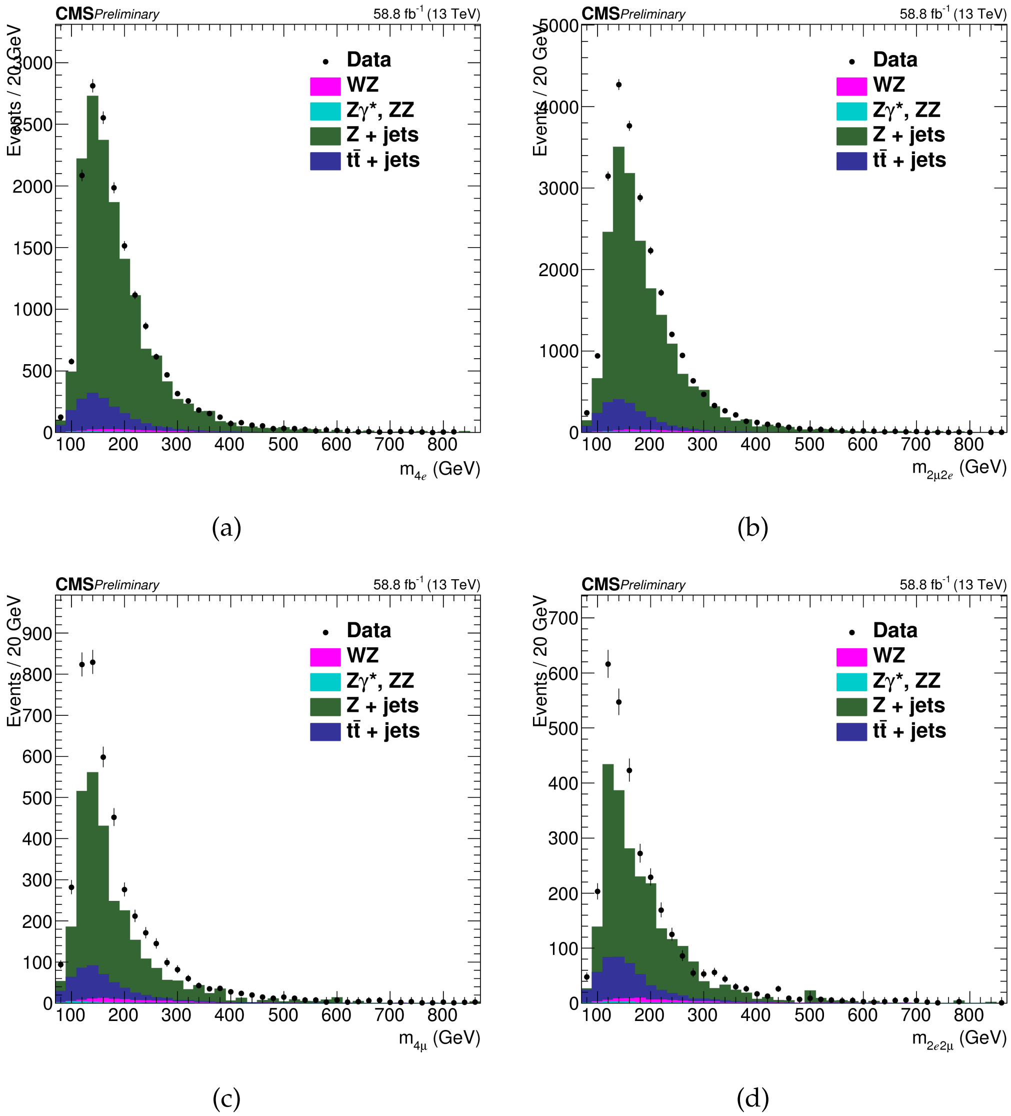


FIGURE 1.17: DATA-MC comparison of the SS-SF samples in the Z+X background control samples. (a) and (b) $4e$ and $2\mu2e$, (c) and (d) 4μ and $2e2\mu$ in 2018 data.

Predictions of the two methods are combined assuming no correlation between the uncertainties (independent control regions, partially different sources systematics), and combined mean values are obtained by weighting the individual means according to the corresponding variances.

Systematic Uncertainties on Reducible Background Source of systematic uncertainty of the methods presented above potentially arises from the different composition of background processes (DY, $t\bar{t}$, WZ, $Z\gamma$) in the region where FRs are

2016	4e	4μ	$2e2\mu$
Method OS	$20.2 \pm 6.2_{\text{tot.}}$	$27.0 \pm 8.6_{\text{tot.}}$	$47.7 \pm 10.5_{\text{tot.}}$
Method SS	$13.1 \pm 5.5_{\text{tot.}}$	$29.6 \pm 9.0_{\text{tot.}}$	$41.5 \pm 10.3_{\text{tot.}}$
Combined	16.2	28.2	44.5
2017	4e	4μ	$2e2\mu$
Method OS	$16.1 \pm 4.9_{\text{tot.}}$	$33.1 \pm 10.4_{\text{tot.}}$	$45.7 \pm 10.2_{\text{tot.}}$
Method SS	$10.9 \pm 4.1_{\text{tot.}}$	$33.4 \pm 10.3_{\text{tot.}}$	$40.8 \pm 9.7_{\text{tot.}}$
Combined	13.0	33.3	43.2
2018	4e	4μ	$2e2\mu$
Method OS	$25.4 \pm 7.7_{\text{tot.}}$	$50.1 \pm 15.5_{\text{tot.}}$	$67.7 \pm 14.8_{\text{tot.}}$
Method SS	$16.1 \pm 5.9_{\text{tot.}}$	$51.9 \pm 15.8_{\text{tot.}}$	$60.7 \pm 14.2_{\text{tot.}}$
Combined	19.4	51.3	64.1

TABLE 1.6: Summary of the results given by the two methods for the prediction of the contribution of reducible background processes in the signal region for all three years. The weighted mean value of the results of the two methods is taken as the final estimate of this contribution, while the uncertainty of the result covers the uncertainties of both methods. The table shows symmetric individual uncertainties for the two methods, while these are treated as asymmetric in the combination and statistical analysis. Results are given for an integrated luminosity of 35.9, 41.5 and 59.7 fb^{-1} in 2016, 2017 and 2018 data, respectively.

measured and are applied. OS method corrects for the resulting bias via the "3P1F component" of its prediction. SS method corrects explicitly the electron fake rates for the correct fraction of photon conversions. The closure tests presented here are used to assess a possible residual bias in the methods.

The limited size of the samples in the control regions where FRS are measured and where those are appleid is the source of the statistical uncertainties of the method. The dominating statistical uncertainty is driven by the number of events in the control region and is typically in the range of 1-10%.

Compositions of reducible background processes (DY, $t\bar{t}$, WZ, $Z\gamma^{(*)}$) in the region where FRS are measured and where those are appleid are typically not the same. This is the main source of the systematic uncertainty of the fake ratio method.

This uncertainty can be estimated by measuring the FRs for individual background processes in the $Z + 1L$ region in simulation. The weighted average of these individual fake ratios is the fake ratio which is measured in this sample (in simulation). The exact composition of the background processes in the 2P+2F region where FRS are to be applied can be determined from simulation, and one can reweigh the individual FRs according to the 2P+2F composition. The difference between the reweighed fake ratio and the average one can be used as a measure of the uncertainty on the measurement of the fake ratios. The effect of this systematic uncertainty is propagated to the final estimates, and it amounts to about 32% for $4e$, 33% for $2e2\mu$ and 35% for 4μ final state.

In order to estimate the uncertainty on the m_{4l} shape, the differences between the shapes of predicted background distributions for all three channels, and between both of the two methods are used. The envelope of differences between these shapes of distributions is used as an estimate of the shape uncertainty. The uncertainty is estimated to be roughly in the range of 5% - 15%. Since the difference of the shapes slowly varies with m_{4l}, it is taken as a constant versus m_{4l} and is absorbed in the much larger uncertainty on the predicted yield of backgrounds.

For differentiall measurements in $\mathrm{H} \to 4\ell$, shape of reducible background is attained from templates built from the control regions in the data for each differential bin of the observable considered for measurement. This procedure is similar to what was used for spin-parity studies as described in the Ref. [32]. More details on this procedure of differential measurements of $\mathrm{H} \to 4\ell$ will be given in Section **??**.

The number of estimated signal and the background events, along with observed events in data passing all selections in mass range $m_{4\ell} > 70$ GeV are listed in Table 1.7 for full Run 2 datasets.

Reconstructed four-leptons invariant mass distribution for whole Run 2 datasets and MC samples in full and high mass range is shownn in the Figure 1.18.

1.4.7 Statistical Procedure

After the procedure of events seletions, signal and background modeling as needed ingredients, next step is to perform final measurements. An asymptotic approach as described in Ref. [37] with test statistics that is based on profile likelihood ratio [38] is used for this purpose. Maximization of likelihood is performed by fitting simultaneously background and signal parametrization to the observed $m_{4\ell}$ distribution, at the fiducial level with all reconstruction bins and all three final states. The parameter of interest (σ_{fid}) is directly extracted from the fit.

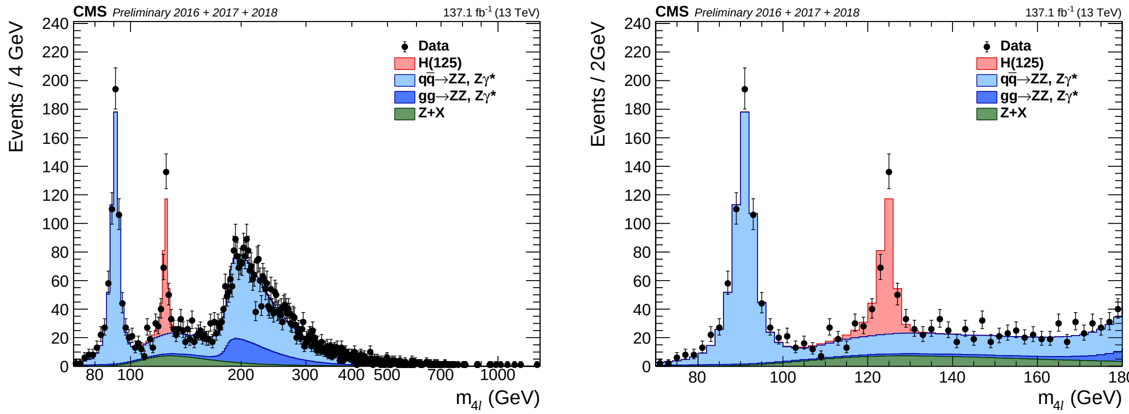


FIGURE 1.18: Invariant mass distributions of invariant mass of four-leptons using full Run 2 datasets in the full (left) and the low (right) mass range, along with predictions for Standard Model Higgs ($m_H = 125.0$ GeV), including $Z \rightarrow 4\ell$ resonance.

The estimated number events for a studied differential observable in a final state f and in the defined bin i are described in terms of $m_{4\ell}$ given by:

$$\begin{aligned} N^{\mathrm{f},i}_{\mathrm{obs}}(m_{4\ell}) &= N^{\mathrm{f},i}_{\mathrm{fid}}(m_{4\ell}) + N^{\mathrm{f},i}_{\mathrm{nonfid}}(m_{4\ell}) + N^{\mathrm{f},i}_{\mathrm{comb}}(m_{4\ell}) + N^{\mathrm{f},i}_{\mathrm{bkg}}(m_{4\ell}) \\ &= \sum_j \epsilon^{\mathrm{f}}_{i,j} \cdot \left(1 + f^{\mathrm{f},i}_{\mathrm{nonfid}}\right) \cdot \sigma^{\mathrm{f},j}_{\mathrm{fid}} \cdot \mathcal{L} \cdot \mathcal{P}_{\mathrm{res}}(m_{4\ell}) \\ &\quad + N^{\mathrm{f},i}_{\mathrm{comb}} \cdot \mathcal{P}_{\mathrm{comb}}(m_{4\ell}) + N^{\mathrm{f},i}_{\mathrm{bkg}} \cdot \mathcal{P}_{\mathrm{bkg}}(m_{4\ell}). \end{aligned} \tag{1.7}$$

Where $N^{\mathrm{f},i}_{\mathrm{fid}}(m_{4\ell})$, $N^{\mathrm{f},i}_{\mathrm{nonfid}}(m_{4\ell})$, $N^{\mathrm{f},i}_{\mathrm{comb}}(m_{4\ell})$ and $N^{\mathrm{f},i}_{\mathrm{bkg}}(m_{4\ell})$ denote the resonant fiducial signal, the nonfiducial resonant, the combinatorial contribution of fiducial signal, and background contributions in a defined bin i as functions of $m_{4\ell}$ (four lepton invariant mass), respectively. Then, probability density functions corresponding to the resonant (fiducial and the nonfiducial) signal, the combinatorial contributions of fiducial signal, and contribution from the background are $\mathcal{P}_{\mathrm{res}}(m_{4\ell})$, $\mathcal{P}_{\mathrm{comb}}(m_{4\ell})$ and $\mathcal{P}_{\mathrm{bkg}}(m_{4\ell})$ respectively. $\mathcal{P}_{\mathrm{res}}(m_{4\ell})$ is described by double sided Crystal Ball function details of which are described in Section 1.4.5.

$\epsilon^{\mathrm{f}}_{i,j}$ is the response matrix that connects reconstructed level events in a defined bin i with fiducial events in the bin j for studied differential observable. It is the probability of an event that was generated in ith bin at the fiducial level and is reconstructed in j-th bin. This is estimated through *unfolding procedure* for observables' bins at the fiducial level, corrected for the detection efficiency and resolution effects. The procedure is discussed in Ref. [39] and similar to what is adopted for

$H \to \gamma\gamma$ decay channel [40]. Figure **??** shows the efficiency matrix for $p_T(H)$ measured using different MC samples for SM production of Higgs. The efficiency matrix is dominated by diagonal elements. In case of jet related observables, off-diagonal contribution, however, is non-negligible. It is due to poor resolution offered by the detector to jets. In case of differential variable, jet multiplicity, neighbour elements of diagonal varies from 3% to 21% as shown in Figures **??**-**??** in the Appendix **??**. Regardless of particular

TABLE 1.7: Number of observed candidates, expected background and the signal events passing all selections in mass range $m_{4\ell} >$ 70 GeV corresponding to three different years. Z+X is estimated by control regions in the data while ZZ and signal events are estimated using MC simulation.

Channel	4e	4μ	$2e2\mu$	4ℓ
2016 (13 TeV)				
$q\bar{q} \to ZZ$	$192.7^{+18.6}_{-20.1}$	$360.2^{+24.9}_{-27.3}$	$471.0^{+32.6}_{-35.7}$	$1023.9^{+68.9}_{-76.0}$
$gg \to ZZ$	$41.2^{+6.3}_{-6.1}$	$69.0^{+9.5}_{-9.0}$	$101.7^{+14.0}_{-13.3}$	$211.8^{+28.9}_{-27.5}$
Z + X	$21.1^{+8.5}_{-10.4}$	$34.4^{+14.5}_{-13.2}$	$59.9^{+27.1}_{-25.0}$	$115.4^{+31.9}_{-30.1}$
Sum of backgrounds	$255.0^{+23.9}_{-25.1}$	$463.5^{+31.9}_{-33.7}$	$632.6^{+44.2}_{-46.1}$	$1351.1^{+85.8}_{-91.2}$
Signal ($m_{\mathrm{H}} = 125$ GeV)	$12.1^{+1.3}_{-1.4}$	23.8 ± 2.1	30.3 ± 2.6	66.3 ± 5.6
Total expected	$267.1^{+24.9}_{-26.1}$	$487.4^{+33.1}_{-34.9}$	$662.9^{+45.7}_{-47.5}$	$1417.4^{+89.1}_{-94.3}$
Observed	293	505	681	1479
2017 (13 TeV)				
$q\bar{q} \to ZZ$	$233.86^{+y.y}_{-z.z}$	$441.08^{+y.y}_{-z.z}$	$569.51^{+y.y}_{-z.z}$	$1244.45^{+y.y}_{-z.z}$
$gg \to ZZ$	$49.11^{+y.y}_{-z.z}$	$81.50^{+y.y}_{-z.z}$	$121.05^{+y.y}_{-z.z}$	$251.66^{+y.y}_{-z.z}$
Z + X	$12.39^{+y.y}_{-z.z}$	$36.39^{+y.y}_{-z.z}$	$44.45^{+y.y}_{-z.z}$	$93.24^{+y.y}_{-z.z}$
Sum of backgrounds	$295.37^{+y.y}_{-z.z}$	$558.97^{+y.y}_{-z.z}$	$735.02^{+y.y}_{-z.z}$	$1589.35^{+y.y}_{-z.z}$
Signal ($m_{\mathrm{H}} = 125$ GeV)	$13.85^{+y.y}_{-z.z}$	$28.64^{+y.y}_{-z.z}$	$35.55^{+y.y}_{-z.z}$	$78.04^{+y.y}_{-z.z}$
Total expected	$309.22^{+y.y}_{-z.z}$	$587.60^{+y.y}_{-z.z}$	$770.57^{+y.y}_{-z.z}$	$1667.39^{+y.y}_{-z.z}$
Observed	307	602	797	1706
2018 (13 TeV)				
$q\bar{q} \to ZZ$	333^{+57}_{-53}	622^{+31}_{-44}	815 ± 73	1770^{+98}_{-101}
$gg \to ZZ$	$75.1^{+14.3}_{-13.5}$	$116.6^{+11.7}_{-12.8}$	176.9 ± 23.0	$368.5^{+29.5}_{-29.6}$
Z + X	19.3 ± 7.2	50.8 ± 15.2	64.6 ± 15.6	134.7 ± 22.9
Sum of backgrounds	$428^{+59.2}_{55.2}$	$790^{+36.4}_{-48.3}$	1057 ± 78.1	$2274^{+104.9}_{-107.7}$
Signal ($m_{\mathrm{H}} = 125$ GeV)	$19.6^{+3.3}_{-3.1}$	$40.8^{+2.5}_{-2.9}$	50.7 ± 5.6	$111.1^{+6.9}_{-7.0}$
Total expected	$447^{+59.3}_{55.2}$	$830^{+36.5}_{-48.4}$	1108 ± 78.3	$2385^{+105.1}_{-107.9}$
Observed	462	850	1130	2442

differential bin, for jet multiplicity observable, the off-diagonal contributions are 3-8%, 3-20% and 2-14%; however, for a high resolution observable, e.g. $p_T(H)$, these are 1-5%, 1-5% and 1-4% for 2e2mu, 4mu and 4e channels respectively in the gluon gluon fusion production mode.

The f^{i}_{nonfid} fraction represents the ratio of nonfiducial signal to the fiducial signal contributions in a defined bin i at reconstruction level. It is the ratio of reconstructed events which are from outside the fiducial volume and reconstructed events which are from within the fiducial volume. Such events arise from imperfect detector resolution which causes the differences for quantities calculated at generated and reconstructed levels. Fraction of these events are also treated as background and will be referred "non-fiducial signal" in the disseration. It has been demonstrated with MC samples that these events can be described using same shape as resonant component. The values of such fraction are extracted from simulation repeatedly for SM. This fraction varies among different production modes of Higgs e.g. it is 4% for the $\mathrm{gg} \to \mathrm{H}$ and 3% for ttH production mode. The values for each model are listed in the Tables **??**, **??** and **??** for 2016, 2017 and 2018 respectively.

$\sigma^{\mathrm{f},j}_{\mathrm{fid}}$ is the parameter of interest that represents signal cross section for a final state f in a defined bin j of fiducial region.

For associate production of Higgs, there is quite a possibility that one reconstructed lepton is not from $\mathrm{H} \to 4\ell$ decay process and orginates associated particle. Fraction of these events, denoted by $\mathcal{P}_{\mathrm{comb}}(m_{4\ell})$, vary with the production modes (i.e. WH, ZH and $t\bar{t}\mathrm{H}$) and are treated as background. They make broad invariant $m_{4\ell}$ mass distribution whose shape is described by a Landau distribution with parameters extracted from simulations.

Main goal is to measure fiducial cross sections after corrections of detector effects, detector resolution and systematic biases. These corrections are extracted with the help of simulated samples and then encoded in the parametrization of $m_{4\ell}$ spectra at the reconstruction level. All the systematic uncertainties in the measurements are encoded as nuisance parameters, which are profited properly in fit procedure.

1.4.8 Extraction of $\mathrm{H} \to 4\ell$ Fiducial Cross Section

Inclusive or Integrated Fiducial Cross section

From measurement point of view, the integrated or inclusive fiducial cross section is a special case of the general procedure adopted for fiducial cross section measurements as explained in previous Section 1.4.7 i.e. the number of considered bins is one.

Informations of different ingredient in Equation 1.7 for different production modes of Higgs are included in the form of tables. Reconstruction efficiency (ϵ) for the events reconstructed in fiducial volume is given in the Table 1.8 for each final state and as well as inclusive for all final states. It is obvious from the table that reconstruction efficiency varies with the production model. Also, in 4μ final state, efficiency is pretty higher that other states which is due to high detector resolution of CMS for muons. Table 1.9 shows the ratio (f_{nonfid}) of events reconstructed from out of fiducial region and the events reconstructed within fiducial volume, per final state. It is evident from the table, f_{nonfid} is higher for $t\bar{t}\mathrm{H}$production mode which is due to the contamination from hadronic activities. Source of the "nonfiducial signal" contributions i.e. f_{nonfid} is the imperfect correspondence of isolation at particle and reconstruction levels, as discussed in previous Section and in Section 1.4. Fraction of the signal events in mass range 105.0–140.0 GeV where at least one selected lepton does not originate from the Higgs boson decay are shown in Table 1.10. Fraction of these events in higher in case of ZH and $t\bar{t}\mathrm{H}$ production modes.

TABLE 1.8: Reconstruction efficiency (ϵ) for fiducial events per final state for different Standard Model signal models at $\sqrt{s} = 13$ TeV with 2018 MC samples.

Sample	$4e$	4μ	$2e2\mu$	4ℓ
gg→H (POWHEG)	0.404 ± 0.002	0.799 ± 0.002	0.572 ± 0.002	0.593 ± 0.001
VBF (POWHEG)	0.431 ± 0.003	0.805 ± 0.002	0.586 ± 0.002	0.608 ± 0.002
WH (POWHEG+MINLO)	0.415 ± 0.004	0.780 ± 0.003	0.580 ± 0.003	0.593 ± 0.002
ZH (POWHEG+MINLO)	0.428 ± 0.006	0.780 ± 0.005	0.581 ± 0.005	0.597 ± 0.003
ttH (POWHEG)	0.429 ± 0.006	0.745 ± 0.005	0.573 ± 0.005	0.585 ± 0.003

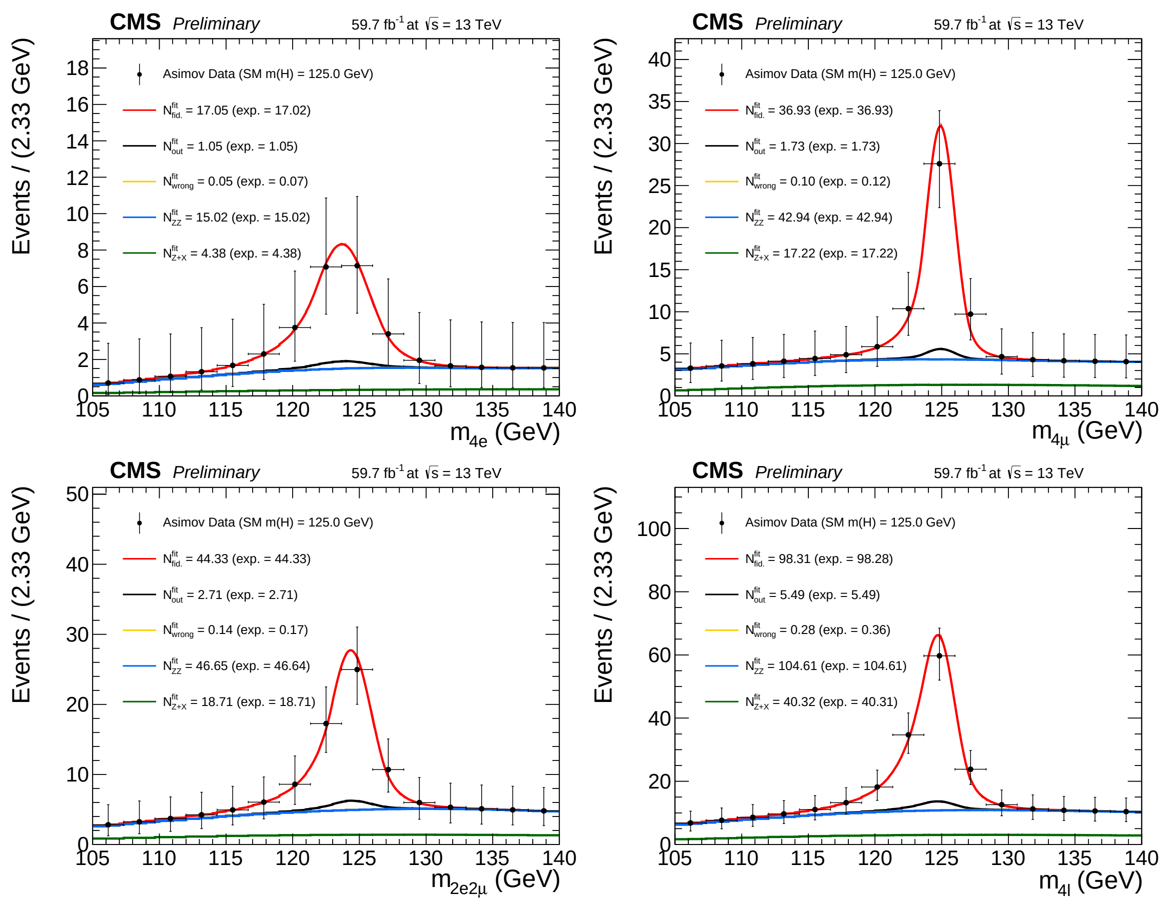


FIGURE 1.19: Inclusive four lepton mass distributions for an Asimov Dataset generated using SM cross section values and SM efficiencies where the gg→H prediction is from POWHEG+JHUGEN and resulting fitted values of the signal and background pdfs in different final states at $\sqrt{s} = 13$ TeV with 2018 data.

TABLE 1.9: Ratio of events reconstructed from out of fiducial volume and the events reconstructed within the fiducial region (f_{nonfid}) per final state in different Standard Model signal models at $\sqrt{s} = 13$ TeV with 2018 MC samples.

Sample	$4e$	4μ	$2e2\mu$	4ℓ
gg→H (POWHEG)	0.060 ± 0.002	0.046 ± 0.001	0.060 ± 0.001	0.055 ± 0.001
VBF (POWHEG)	0.050 ± 0.002	0.037 ± 0.001	0.046 ± 0.001	0.043 ± 0.001
WH (POWHEG+MINLO)	0.098 ± 0.004	0.058 ± 0.002	0.080 ± 0.002	0.075 ± 0.001
ZH (POWHEG+MINLO)	0.105 ± 0.007	0.071 ± 0.004	0.091 ± 0.004	0.087 ± 0.002
ttH (POWHEG)	0.278 ± 0.012	0.127 ± 0.005	0.193 ± 0.006	0.185 ± 0.004

TABLE 1.10: Signal event fraction in the mass range 105.0–140.0 GeV where minimum one lepton selected is originating from Higgs decay at $\sqrt{s} = 13$ TeV with 2018 MC samples.

Sample	$4e$	4μ	$2e2\mu$	4ℓ
gg→H (POWHEG)	0.002	0.002	0.002	0.002
VBF (POWHEG)	0.002	0.003	0.003	0.003
WH (POWHEG+MINLO)	0.036	0.031	0.042	0.037
ZH (POWHEG+MINLO)	0.199	0.207	0.212	0.208
ttH (POWHEG)	0.175	0.156	0.164	0.163

Differential Fiducial Cross section

In this dissertation, the kinematic properties of four-lepton system are exploited for Higgs differential cross section measurements and the results are reported for transverse momentum and rapidity of Higgs or the four lepton system($p_T(\mathrm{H})$ or $p_T(4\ell)$, $|y(\mathrm{H})|$ or $|y(4\ell)|$), N(jets), and $p_T(\mathrm{jet})$ which probe the production mechanism. For each kinematic or differential observable, selection procedure of the binning boundaries has been optimized in order to have a similar expected measured cross-section for every bin of the observable.

For the differential cross section measurements, the background is to be subtracted at the reconstruction level for each bin of the observable. The mass spectrum for the background in not same all differential bins because it does not have a resonance structure. The mass spectrum can also have non-negligible correlation with the observables.

For the irreducible backgrounds, the four-lepton mass spectra are modeled as the $m_{4\ell}$ template shapes through simulated events for each bin of the observable for differential measurements. However, for the reducible backgrounds the templates are built from the control regions in the data in the same way. The procedure is described in Ref. [32].

In each differential bin of $p_T(4\ell)$ fractions of background events are presented in the Table 1.11.

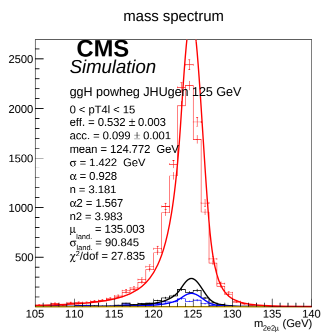
TABLE 1.11: Estimated fraction of background events in each bin of $p_T(4\ell)$, for each final state using 2018 samples.

Background	final state	$0-15$	$15-30$	$30-45$	$45-80$	$80-120$	$120-200$	$200-13000$
qqZZ	$2e2\mu$	0.483668	0.239196	0.130151	0.105276	0.0266332	0.011809	0.00326633
qqZZ	$4e$	0.459665	0.234399	0.136986	0.114916	0.0388128	0.00989346	0.00532725
qqZZ	4μ	0.492632	0.241203	0.110977	0.109173	0.0297744	0.0129323	0.00330827
ggZZ	$2e2\mu$	0.310446	0.309859	0.191901	0.168134	0.0187793	0.000880282	0.0
ggZZ	$4e$	0.3085	0.306332	0.19247	0.17433	0.0169994	0.00136908	0.0
ggZZ	4μ	0.321332	0.319311	0.194863	0.150987	0.0128703	0.000638196	0.0
Z+X (CR)	4l	0.0992521	0.211153	0.209307	0.305327	0.120303	0.0483797	0.00627828

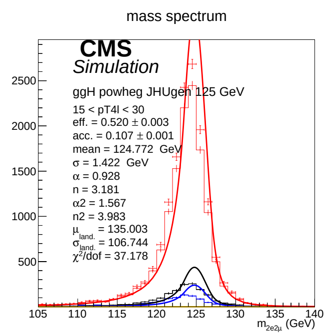
The fiducial differential cross section is extracted in the bins of kinematic observables at the fiducial level following the procedure outlined in Section 1.4.7. For signal yields, mass spectrum of signal is built in each cell of the response matrix using a Double Crytal Ball (DCB) function, similar to what was done in Ref. [32]. In practice, the same parameters of the CB function are used in each cell and the systematic uncertainties assigned on the energy scale and resolution generally cover discrepancy of signal mass spectrum among each cell.

Examples of the fitted signal line shapes for differential bins of $p_T(\mathrm{H})$ from gg→H (POWHEG+JHUGEN 125 GeV 2018 sample) in the individual final states are shown in Figure 1.20, and examples of the efficiency matrices for $p_T(\mathrm{H})$ for the $2e2\mu$ final state are shown in Fig 1.21. Many more examples can be seen in Appendix **??**.

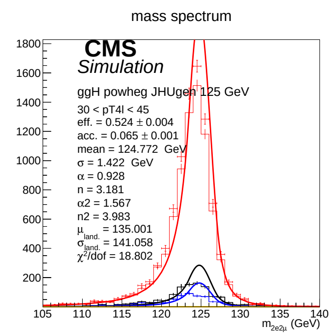
Figure 1.22 shows the four lepton mass distributions for an Asimov dataset generated using SM cross section values and efficiencies from the gg→H production mode from POWHEG+JHUGEN, and resulting fitted values of PDFs of signal and background for different bins of $p_T(\mathrm{H})$ (all final states combined).



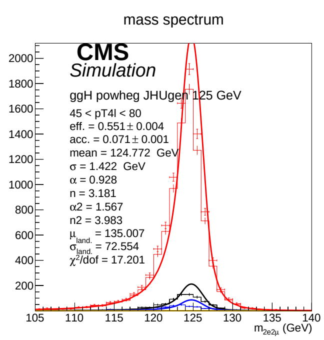
(a) $0.0 < p_T(\mathrm{H}) < 15.0$



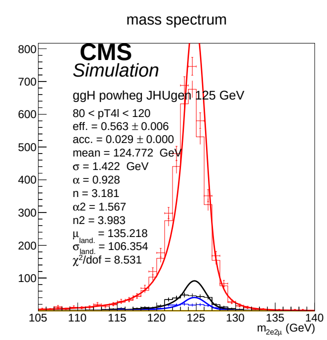
(b) $15.0 < p_T(\mathrm{H}) < 30.0$



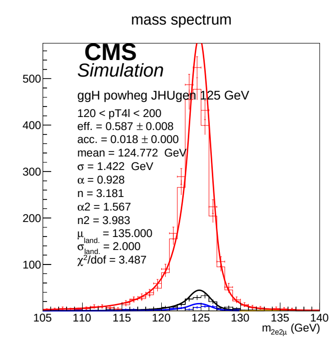
(c) $30.0 < p_T(\mathrm{H}) < 45.0$



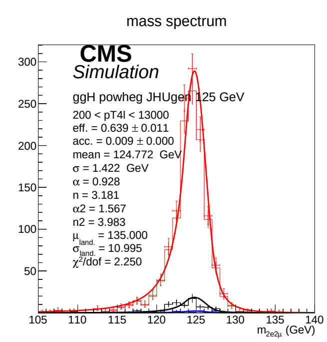
(d) $45.0 < p_T(\mathrm{H}) < 80.0$



(e) $80.0 < p_T(\mathrm{H}) < 120.0$

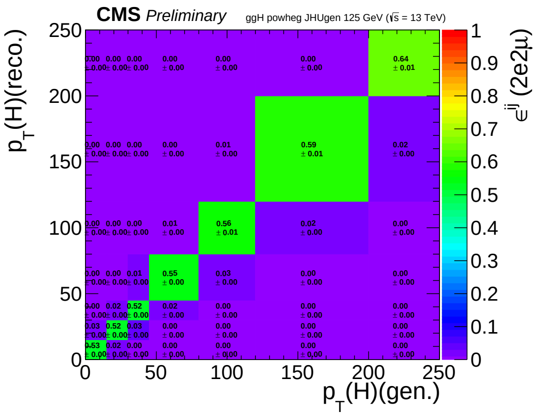


(f) $120.0 < p_T(\mathrm{H}) < 200.0$

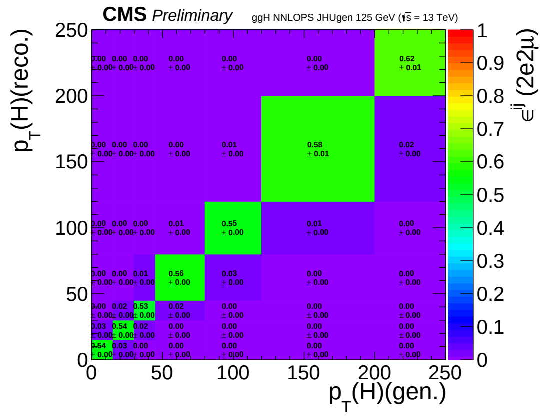


(g) $200.0 < p_T(\mathrm{H}) < 13000.0$

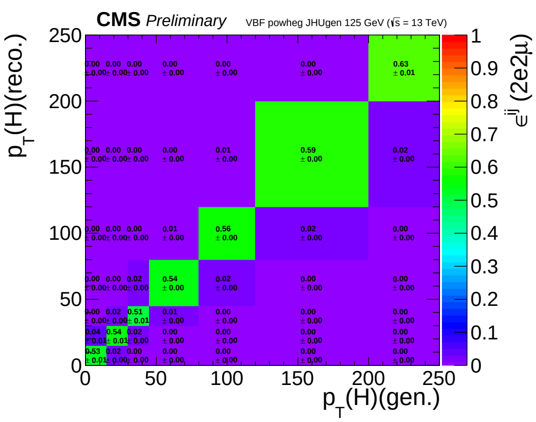
FIGURE 1.20: Example signal shapes at reconstruction level for a resonance of m(4ℓ) in $2e2\mu$ final state for the $gg \to$ H production mode from POWHEG+JHUGEN in different bins of $p_T(\mathrm{H})$. The black curve represents events which do not pass the fiducial volume selection. The curve has no effect on the result.



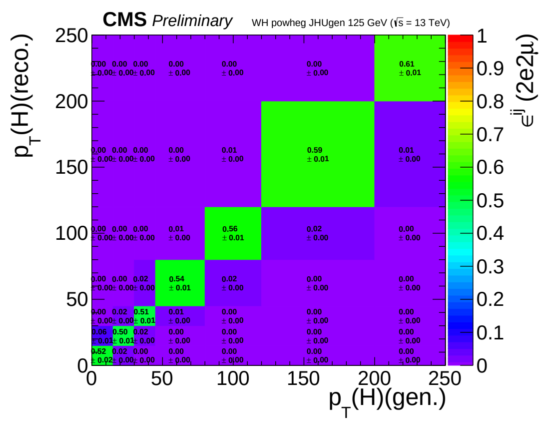
(a) $gg \to$H (POWHEG+JHUGEN)



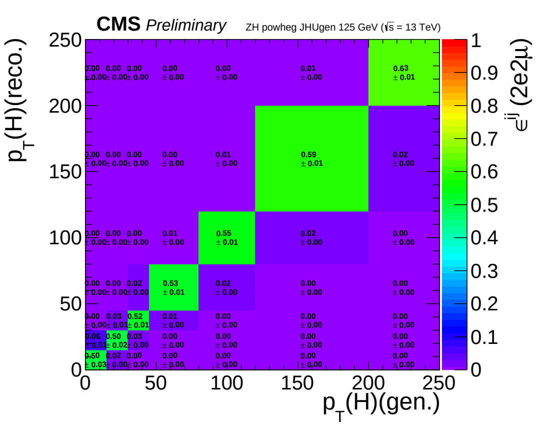
(b) $gg \to$H (MINLOHJJ)



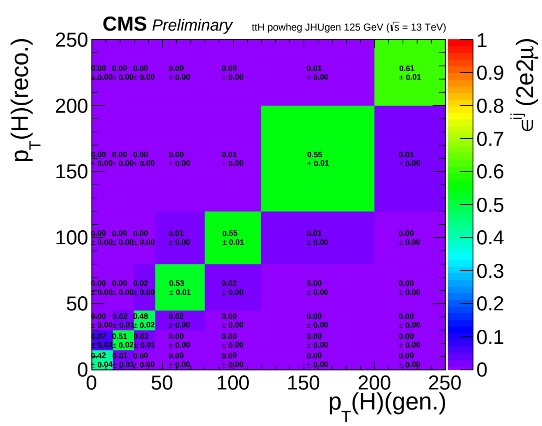
(c) VBF (POWHEG)



(d) WH (PYTHIA)



(e) ZH (PYTHIA)



(f) ttH (PYTHIA)

FIGURE 1.21: Efficiency response matrices for p_T(H) for different SM production modes in the $2e2\mu$ final state using 2018 samples.

1.5 Systematic Uncertainties

1.5.1 Experimental uncertainties souces

Main experimental uncertainties affecting signal and background are uncertainty on integrated luminosity (from 2.3% to 2.6%, depending on the year of data taking) and the uncertainty on efficiencies of lepton reconstruction and identification (ranging from 2.5% (4μ)–16.1% ($4e$)). Uncertainties on estimation of reducible backround, described in Section 1.4.6, originating from the background composition and misidentification rate uncertainty vary for different final states. Mismatch in composition of backgrounds between different samples where the fake rate is derived and where it is applied accounts between 24%(4μ) and 43%(($4e$)).

Lepton energy scale uncertainty is assessed by propagating down to the four-leptons invariant mass the uncertainty associated the lepton momentum corrections on each indiviudal lepton. It is estimated by exploiting $Z \to \ell\ell$ invariant mass distributions both in data and the simulation. Events are categorized on the basis p_T and η of either (randomly determined) of two leptons and integrating on the other lepton. The dilepton mass distributions are then fit by parameterization of a Breit-Wigner convoluted with a DCB function. The uncertainty is treated as uncorrelated (all up or all down). The new four-leptons invariant mass obtained this way is then fitted with only the mean value floating. Difference between new mean value and default one gives an estimation of impact of lepton energy scale. Values of the differences are reported in the Table 1.12, together with estimation of the impact of lepton momentum resolution, following a similar procedure.

TABLE 1.12: Difference (in %) between the nominal mean and the width of four-leptons or Higgs invariant mass distribution and those where the lepton momentum scale and resolution uncertainties have been propagated.

	4e	4μ	$2e2\mu$
Scale Difference (%)	0.087	0.049	0.069
Resolution Difference (%)	17	12	17

The results obtained this way suggest that the uncertainties from 2016 (20% on the resolution, 0.3% for the $4e$ channel for scale) are still valid.

1.5.2 Theoretical uncertainties

Estimation of the theoretical uncertainties affecting signal and the background include uncertainties from factorization and renormalization scale is done by varying the two scales 0.5-2 times the nominal values with their ratio kept in between 0.5-2. In the case of ggF production mode, the QCD scale uncertainty is broke down to 9 different sources. Other theoretical uncertainty source is PDF set choice which is determined RMS of the variation when different replias of default NNPDF set is used. Additional uncertainty on the background which is 10% on K factor applied on $ggZZ$ prediction is considered as described in Section 1.4.6. Systematic uncertainty of about 2% on the branching ratio of $\mathrm{H} \rightarrow 4\ell$ is only applicable to signal yield. Additional sources are the imprecise jet energy scale and resolution knowledge (from 2% to 22%), the efficiency on b-tagging and the light quarks ($i.e.$ u, d, s, c) and gluon jet mistag rate. While cross section measurement, signal cross section uncertainties arising from theoretical sources are removed.

TABLE 1.13: Summary of the experimental systematic uncertainties in the $\mathrm{H} \rightarrow 4\ell$ measurements of 2016, 2017 and 2018 data.

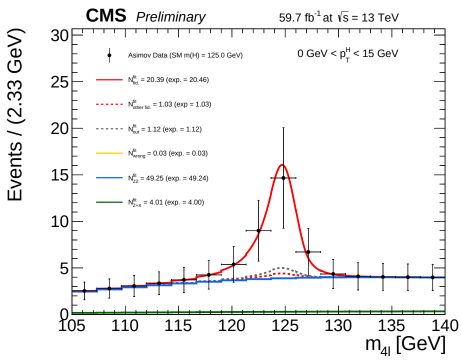
Summary of relative systematic uncertainties			
Common experimental uncertainties			
	2016	2017	2018
Luminosity	2.6 %	2.3 %	2.5 %
Lepton identification/reconstruction efficiencies	2.5 – 9 %	3 – 12.5 %	0.7 – 11 %
Background related uncertainties			
Reducible background (Z+X)	36 – 43 %	6 – 32 %	31 –37 %
Signal related uncertainties			
Lepton energy scale	0.04 – 0.3 %	0.04 – 0.3 %	0.04 – 0.3 %
Lepton energy resolution	20 %	20 %	20 %

TABLE 1.14: Table theory uncertainties in the $\mathrm{H} \rightarrow 4\ell$ Inclusive cross section measurements

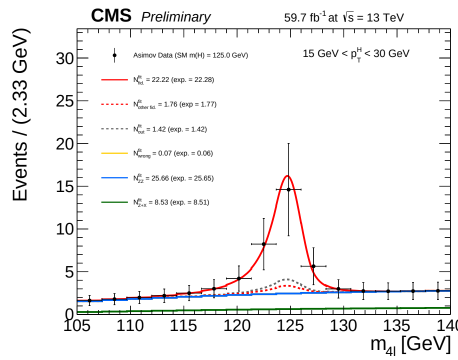
Summary of inclusive theory uncertainties	
BR($H \rightarrow$ ZZ $\rightarrow 4\ell$)	2 %
QCD scale (qq $\rightarrow$ ZZ)	+3.2/-4.2 % %
PDF set (qq $\rightarrow$ ZZ)	+3.1/-3.4 %
Electroweak corrections (qq $\rightarrow$ ZZ)	$\pm$ 0.1 %

1.5.3 Systematic uncertainties treatment when combining yearly measurements on data

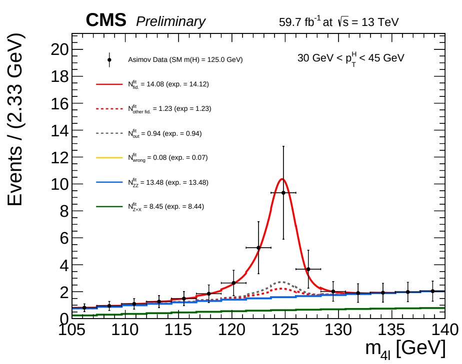
In combination of three data taking periods, theoretical uncertainties as well as the experimental ones related to leptons or jets are treated as correlated while all other ones from experimental sources are taken as uncorrelated.



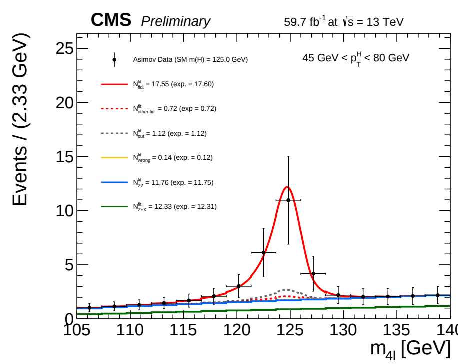
(a) 0.0 GeV $< p_T(\mathrm{H}) <$ 15.0 GeV



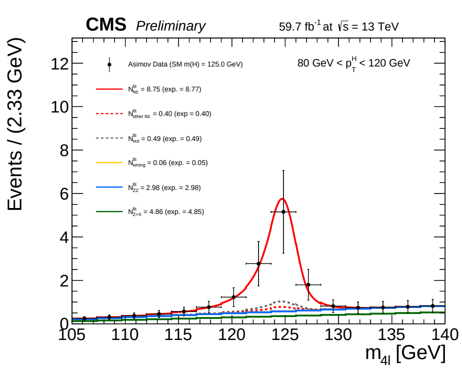
(b) 15.0 GeV $< p_T(\mathrm{H}) <$ 30.0 GeV



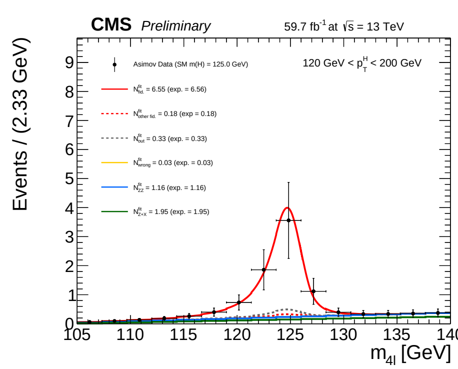
(c) 30.0 GeV $< p_T(\mathrm{H}) <$ 45.0 GeV



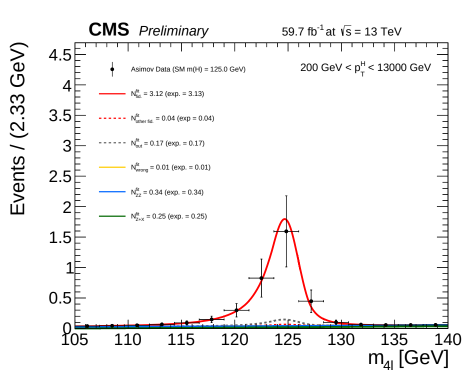
(d) 45.0 GeV $< p_T(\mathrm{H}) <$ 80.0 GeV



(e) 80.0 GeV $< p_T(\mathrm{H}) <$ 120.0 GeV



(f) 120.0 GeV $< p_T(\mathrm{H}) <$ 200.0 GeV



(g) 200.0 GeV $< p_T(\mathrm{H}) <$ 13000.0 GeV

FIGURE 1.22: Four lepton mass distributions for an Asimov dataset generated using SM cross section values and SM efficiencies where the gg→H prediction is from POWHEG+JHUGEN and resulting fitted values of PDFs of signal and background for different bins of $p_T(\mathrm{H})$ (all final states combined).

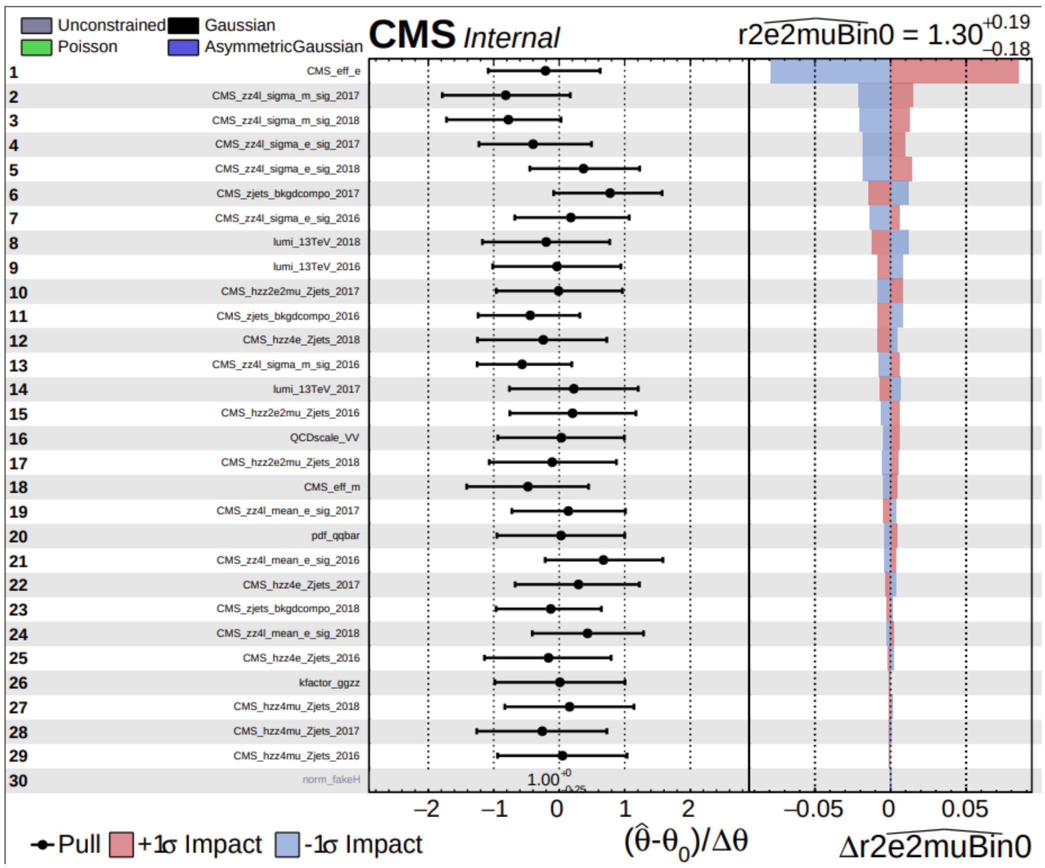


FIGURE 1.23: Pulls. Overall impact of different uncertainties sources on the inclusive measurement ($2e2\mu$ final state)

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